

final_figs_tables_PC copy

September 14, 2026

1 Load packages

```
[1]: import scanpy as sc
import anndata as ad
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
#import scselpy as scS
from gseapy.plot import gseaplot
import gseapy as gp
from gseapy import dotplot
plt.rcParams['pdf.fonttype']=42
plt.rcParams["font.family"] = "Helvetica"
```

2 Load Data

```
[2]: integrated = sc.read_h5ad("/Users/chittiboinap/Library/CloudStorage/
↳OneDrive-NationalInstitutesofHealth/NIH Research/NF2/nf2 schwannoma paper/
↳data/Ihika_Analysis/results/integrated/integrated_new/integrated_GEX.h5ad")
```

```
[9]: schwann = sc.read_h5ad('/Users/chittiboinap/Library/CloudStorage/
↳OneDrive-NationalInstitutesofHealth/NIH Research/NF2/nf2 schwannoma paper/
↳data/Ihika_Analysis/results/integrated/integrated_new/Schwann_GEX.h5ad')
```

2.1 Create new MDS

```
[6]: new_mds = ['ADARB1', 'AKT3', 'ANTXR2', 'CDC42BPA', 'CDH2', 'CERS6', 'CHIC2',
↳'CMC1', 'CMSS1',
'COL12A1', 'COL5A2', 'CSNK1G3', 'DAAM1', 'DLG1', 'DNAJC1', 'DNAJC6',
↳'EPHA3', 'FAM126A',
'FBXL2', 'FBXL5', 'FGF2', 'FHOD3', 'GALNT18', 'GLIS3', 'GRB14',
↳'ITGA1', 'LINCO0968', 'LPAR1',
'MAP2', 'MAP3K20', 'MEG8', 'MINDY3', 'MIR222HG', 'MPDZ', 'MRPL1',
↳'NAV3', 'NBAS', 'NCKAP1', 'PCDH9',
'PDE5A', 'PLSCR4', 'PPP2R2A', 'PREP', 'PTPN12', 'RAP1GDS1',
↳'SAMD12', 'SCFD1', 'SH3D19', 'SIPA1L1',
```

```

        'SLIT2', 'SOS1', 'SUPT3H', 'TBCK', 'TMEM267', 'TSPAN5', 'UBE3A',
        ↪ 'UBR1', 'VPS13A', 'WDR70', 'ZC3H12C', 'ZDHHC20', 'ZDHHC21']
sc.tl.score_genes(integrated, gene_list=new_mds, score_name='new_mds',
        ↪ use_raw=False)

```

OMP: Info #276: omp_set_nested routine deprecated, please use
omp_set_max_active_levels instead.

3 Aesthetics

```

[3]: colors = [
    "#3498db", # Royal Blue
    "#e74c3c", # Alizarin Crimson
    "#2ecc71", # Emerald
    "#f39c12", # Sunflower
    "#1abc9c", # Turquoise
    "#9b59b6", # Amethyst
    "#e67e22", # Carrot
    "#27ae60", # Nephritis
    "#c0392b"  # Pomegranate
]

# ct_dict = {}
# for i in range(0, len(cell_types)):
#     ct_dict[cell_types[i]] = colors[i]

# schwann_dict = {}
# for i in range(0, len(schwann_subclasses)):
#     schwann_dict[schwann_subclasses[i]] = colors[i]

# macro_dict = {}
# for i in range(0, len(macrophage_subclasses)):
#     macrophage_dict[macrophage_subclasses[i]] = colors[i]

# NF2_spor_dict = {'NF2': colors[3], 'Sporadic': colors[0]}

# Hard code colors so it looks better
ct_dict = {
    'Macrophage': "#3498db",
    'Endothelial': "#e74c3c",
    'T-cells': "#2ecc71",
    'Fibroblast': "#f39c12",
    'Unknown': "#1abc9c",
    'Schwann': "#9b59b6",
    'Pericytes': "#e67e22",
    'Neutrophil': "#27ae60",
    'NK Cells': "#c0392b"
}

```

```

}

schwann_dict = {
    'Macrophage-like':"#3498db",
    'Endothelial-like':"#e74c3c",
    'IFN_Response':"#2ecc71",
    'VEGFA+':"#f39c12",
    'Repair-like':"#1abc9c",
    'Classical':"#9b59b6",
    'Proliferating':"#27ae60",
    'Myelinating':"#c0392b"
}

macro_dict = {
    'M1-like':"#3498db",
    'M2-like':"#e74c3c",
    'Pro-Inflammatory':"#f39c12",
    'Regulatory':"#9b59b6",
    'Proliferating':"#27ae60"
}

macro_dict1 = {
    'Macrophage_M1-like':"#3498db",
    'Macrophage_M2-like':"#e74c3c",
    'Macrophage_Pro-Inflammatory':"#f39c12",
    'Macrophage_Regulatory':"#9b59b6",
    'Macrophage_Proliferating':"#27ae60"
}

NF2_spor_dict = {'NF2':"#e74c3c", 'Sporadic':"#3498db"}

# Samples per genotype:
#NF2 = scVS01, scVS02, snVS01, snVS02, snVS03, snVS04, snVS05, snVS06, snVS07,
↪snVS08, snVS09
#Sporadic = scVS10, scVS11, snVS10, snVS11, snVS12, snVS13, snVS14, snVS15

# Define color palettes for NF2 and Sporadic
nf2_colors = ['#e74c3c', '#d35400', '#c0392b', '#e67e22', '#f39c12',
              '#cd6155', '#a93226', '#ba4a00', '#f1c40f', '#d68910', '#b9770e']
↪ # 11 colors for NF2

sporadic_colors = ['#3498db', '#2980b9', '#1abc9c', '#2ecc71', '#16a085',
                  '#8e44ad', '#34495e', '#5dade2'] # 8 colors for Sporadic

# NF2 'orig.ident's'
nf2_orig_ids = ['scVS01', 'scVS02', 'snVS01', 'snVS02', 'snVS03',
                'snVS04', 'snVS05', 'snVS06', 'snVS07', 'snVS08', 'snVS09']

```

```

# Sporadic 'orig.ident'
sporadic_orig_ids = ['scVS10', 'scVS11', 'snVS10', 'snVS11', 'snVS12',
                    'snVS13', 'snVS14', 'snVS15']

# Create color dictionary
color_dict = {}

for orig_ident, color in zip(nf2_orig_ids, nf2_colors):
    color_dict[orig_ident] = color

for orig_ident, color in zip(sporadic_orig_ids, sporadic_colors):
    color_dict[orig_ident] = color

# Print the color dictionary
print(color_dict)

dict_22q = {'22q+/+': '#e74c3c', '22q+/-': '#90db34', '22q-/-': '#3498db', 'NA':
            '↪#d3d3d3'}

```

```

{'scVS01': '#e74c3c', 'scVS02': '#d35400', 'snVS01': '#c0392b', 'snVS02':
'#e67e22', 'snVS03': '#f39c12', 'snVS04': '#cd6155', 'snVS05': '#a93226',
'snVS06': '#ba4a00', 'snVS07': '#f1c40f', 'snVS08': '#d68910', 'snVS09':
'#b9770e', 'scVS10': '#3498db', 'scVS11': '#2980b9', 'snVS10': '#1abc9c',
'snVS11': '#2ecc71', 'snVS12': '#16a085', 'snVS13': '#8e44ad', 'snVS14':
'#34495e', 'snVS15': '#5dade2'}

```

4 Create subsets

```

[ ]: ## Create subsets for each sample prep and type
sc_NF2 = integrated[(integrated.obs['type']=='NF2') & (integrated.obs['orig.
↪ident'].isin(['scVS01', 'scVS02', 'scVS10', 'scVS11']))]
sc_Spor = integrated[(integrated.obs['type']=='Sporadic') & (integrated.
↪obs['orig.ident'].isin(['scVS01', 'scVS02', 'scVS10', 'scVS11']))]
sn_NF2 = integrated[(integrated.obs['type']=='NF2') & (integrated.obs['orig.
↪ident'].isin(['snVS01', 'snVS02', 'snVS03', 'snVS04', 'snVS05', 'snVS06',
↪'snVS07', 'snVS08', 'snVS09', 'snVS10', 'snVS11', 'snVS12', 'snVS13',
↪'snVS14', 'snVS15']))]
sn_Spor = integrated[(integrated.obs['type']=='Sporadic') & (integrated.
↪obs['orig.ident'].isin(['snVS01', 'snVS02', 'snVS03', 'snVS04', 'snVS05',
↪'snVS06', 'snVS07', 'snVS08', 'snVS09', 'snVS10', 'snVS11', 'snVS12',
↪'snVS13', 'snVS14', 'snVS15']))]

```

5 Figure 1 Study overview

5.1 Figure 1C

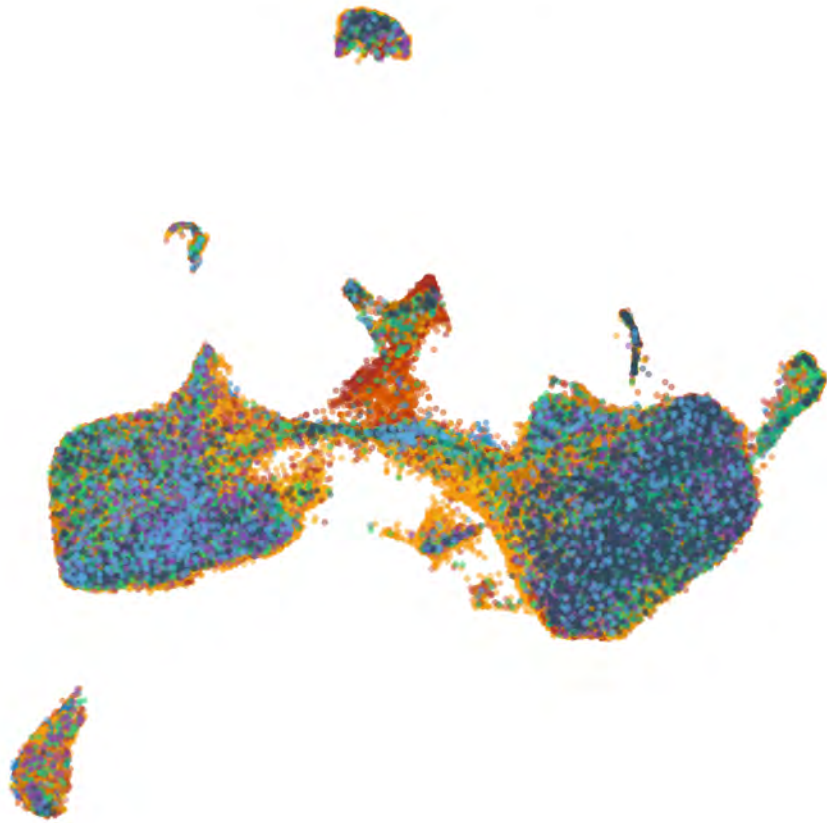
```
[35]: ## Plot overall Integrated object colored by sample
fig, ax1 = plt.subplots(nrows=1,ncols=1,figsize=(6,6), dpi=600)
ax1_dict = sc.pl.umap(integrated,
                      use_raw = False,
                      frameon=False,
                      show=False,
                      ax = ax1,
                      color = "orig.ident",
                      s = 25,
                      palette = color_dict,
                      # edgecolor = 'black',
                      # linewidth= 0.5,
                      alpha = 0.5,
                      )

# handles, labels = plt.gca().get_legend_handles_labels()

# leg1 = ax1.legend(prop = {'size':'14'}, markerscale=1.6, bbox_to_anchor=(0,1),
↪ 1))
# for lh in leg1.legend_handles:
#     lh.set_alpha(0.9)
#     lh.set_edgecolor("black")
ax1.get_legend().remove()
ax1.set_title("Samples",fontweight="bold", fontsize=24)
fig.savefig('figure 1/samples_noedge.pdf',bbox_inches='tight')
fig.savefig('figure 1/samples_noedge.png',bbox_inches='tight')

/Users/mullaneydb/anaconda3/lib/python3.9/site-
packages/scanpy/plotting/_tools/scatterplots.py:392: UserWarning: No data for
colormapping provided via 'c'. Parameters 'cmap' will be ignored
    cax = scatter(
```

Samples



6 Single cell vs nuclei

```
[48]: ## VS01 + VS10 cells and nuclei

# Subset the datasets with parallel cell and nucleus data
cell_vs_nuc = integrated[(integrated.obs['orig.ident'].
    ↳isin(['scVS01', 'scVS02', 'snVS01', 'snVS02',
        'scVS10', 'scVS11', 'snVS10', 'snVS11']))]

# Subset single samples for plotting
vs01cell = cell_vs_nuc[(cell_vs_nuc.obs['orig.ident'].isin(['scVS01']))]
vs01nuc = cell_vs_nuc[(cell_vs_nuc.obs['orig.ident'].isin(['snVS01']))]
vs10cell = cell_vs_nuc[(cell_vs_nuc.obs['orig.ident'].isin(['scVS10']))]
vs10nuc = cell_vs_nuc[(cell_vs_nuc.obs['orig.ident'].isin(['snVS10']))]
```

```

[49]: ## Plot cells/nuclei on separate UMAP projections for each prep and type

fig, ((ax1, ax2), (ax3, ax4)) = plt.subplots(nrows=2,ncols=2,figsize=(6,6),
↳dpi=600)
ax1_grey = sc.pl.umap(integrated,
                      use_raw = False,
                      frameon=False,
                      show=False,
                      ax = ax1,
                      # color = "pred.type",
                      s = 30,
                      # palette = ct_dict,
                      # edgecolor = 'black',
                      # linewidth= 0.5,
                      alpha = 0.9,
                      )
ax1_dict = sc.pl.umap(vs01cell,
                      use_raw = False,
                      frameon=False,
                      show=False,
                      ax = ax1,
                      color = "pred.type",
                      s = 30,
                      palette = ct_dict,
                      edgecolor = 'black',
                      linewidth= 0.5,
                      alpha = 0.9,
                      )
ax2_grey = sc.pl.umap(integrated,
                      use_raw = False,
                      frameon=False,
                      show=False,
                      ax = ax2,
                      # color = "pred.type",
                      s = 30,
                      # palette = ct_dict,
                      # edgecolor = 'black',
                      # linewidth= 0.5,
                      alpha = 0.9,
                      )
ax2_dict = sc.pl.umap(vs01nuc,
                      use_raw = False,
                      frameon=False,
                      show=False,
                      ax = ax2,
                      color = "pred.type",
                      s = 25,

```

```

        palette = ct_dict,
        edgecolor = 'black',
        linewidth= 0.5,
        alpha = 0.9,
    )
ax3_grey = sc.pl.umap(integrated,
    use_raw = False,
    frameon=False,
    show=False,
    ax = ax3,
    # color = "pred.type",
    s = 30,
    # palette = ct_dict,
    # edgecolor = 'black',
    # linewidth= 0.5,
    alpha = 0.9,
)
ax3_dict = sc.pl.umap(vs10cell,
    use_raw = False,
    frameon=False,
    show=False,
    ax = ax3,
    color = "pred.type",
    s = 30,
    palette = ct_dict,
    edgecolor = 'black',
    linewidth= 0.5,
    alpha = 0.9,
)
ax4_grey = sc.pl.umap(integrated,
    use_raw = False,
    frameon=False,
    show=False,
    ax = ax4,
    # color = "pred.type",
    s = 30,
    # palette = ct_dict,
    # edgecolor = 'black',
    # linewidth= 0.5,
    alpha = 0.9,
)
ax4_dict = sc.pl.umap(vs10nuc,
    use_raw = False,
    frameon=False,
    show=False,
    ax = ax4,
    color = "pred.type",

```

```

        s = 25,
        palette = ct_dict,
        edgecolor = 'black',
        linewidth= 0.5,
        alpha = 0.9,
    )

#add legend to plot
handles, labels = plt.gca().get_legend_handles_labels()

leg1 = ax1.legend(prop = {'size':'10'}, markerscale=1.2, bbox_to_anchor=(0, 1))
for lh in leg1.legend_handles:
    lh.set_alpha(0.9)
    lh.set_edgecolor("black")

ax2.get_legend().remove()
ax3.get_legend().remove()
ax4.get_legend().remove()

ax1.set_title("VS01 Cells", fontweight="bold", fontsize=16)
ax2.set_title("VS01 Nuclei ", fontweight="bold", fontsize=16)
ax3.set_title("VS10 Cells", fontweight="bold", fontsize=16)
ax4.set_title("VS10 Nuclei", fontweight="bold", fontsize=16)

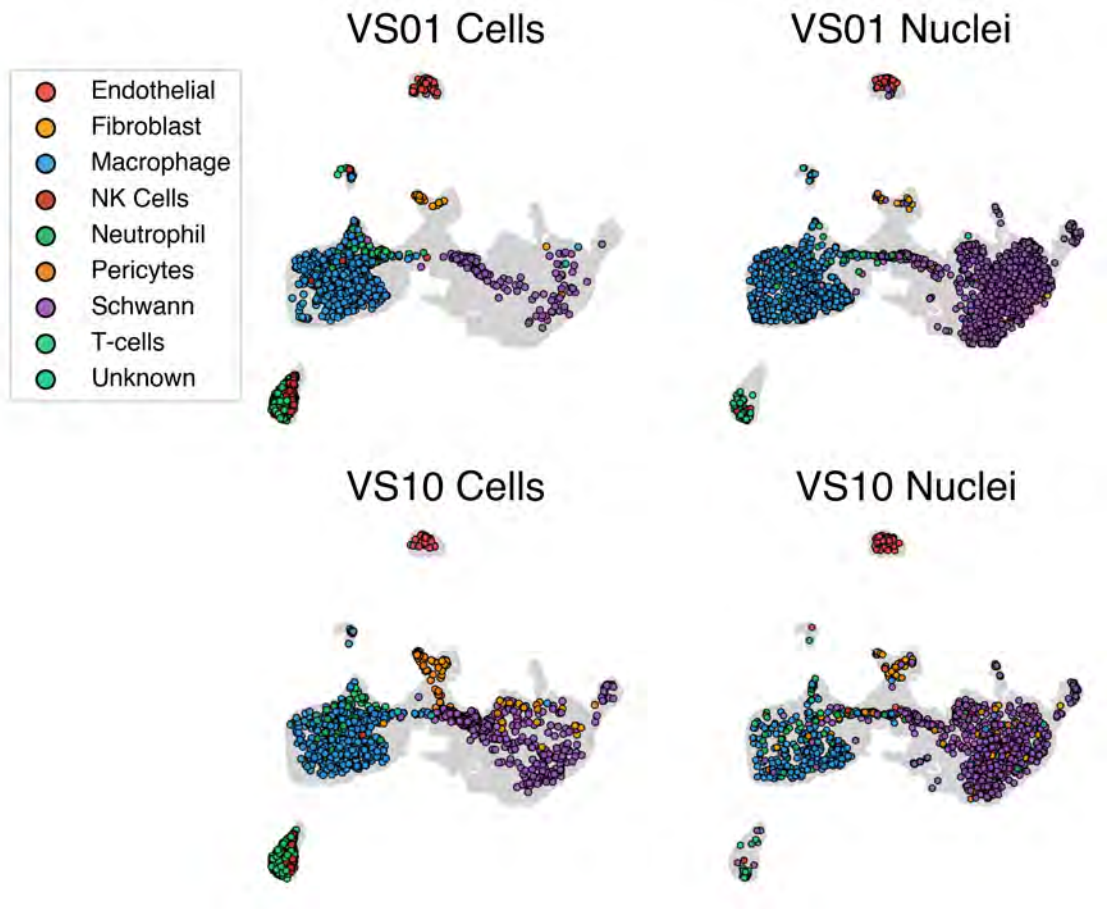
fig.savefig('figure 1/cells_vs_nuclei.pdf',bbox_inches='tight')
fig.savefig('figure 1/cells_vs_nuclei.png',bbox_inches='tight')

```

```

/Users/chittiboinap/anaconda3/envs/singlecell/lib/python3.13/site-
packages/scanpy/plotting/_utils.py:482: ImplicitModificationWarning: Trying to
modify attribute `._uns` of view, initializing view as actual.
    adata.uns[value_to_plot + "_colors"] = colors_list
/Users/chittiboinap/anaconda3/envs/singlecell/lib/python3.13/site-
packages/scanpy/plotting/_utils.py:482: ImplicitModificationWarning: Trying to
modify attribute `._uns` of view, initializing view as actual.
    adata.uns[value_to_plot + "_colors"] = colors_list
/Users/chittiboinap/anaconda3/envs/singlecell/lib/python3.13/site-
packages/scanpy/plotting/_utils.py:482: ImplicitModificationWarning: Trying to
modify attribute `._uns` of view, initializing view as actual.
    adata.uns[value_to_plot + "_colors"] = colors_list
/Users/chittiboinap/anaconda3/envs/singlecell/lib/python3.13/site-
packages/scanpy/plotting/_utils.py:482: ImplicitModificationWarning: Trying to
modify attribute `._uns` of view, initializing view as actual.
    adata.uns[value_to_plot + "_colors"] = colors_list
1 extra bytes in post.stringData array
'created' timestamp seems very low; regarding as unix timestamp

```



```
[71]: from gseapy import dotplot
# to save your figure, make sure that ``ofname`` is not None
dotplot(pre_res.res2d,
        column="FDR q-val",
        title='Pathways enriched in single nucleus samples',
        cmap=plt.cm.viridis,
        size=4, # adjust dot size
        figsize=(4,6), cutoff=0.61, show_ring=False,
        ofname='sup fig s1/nuclei_enriched_pathways_dotplot.pdf')
```

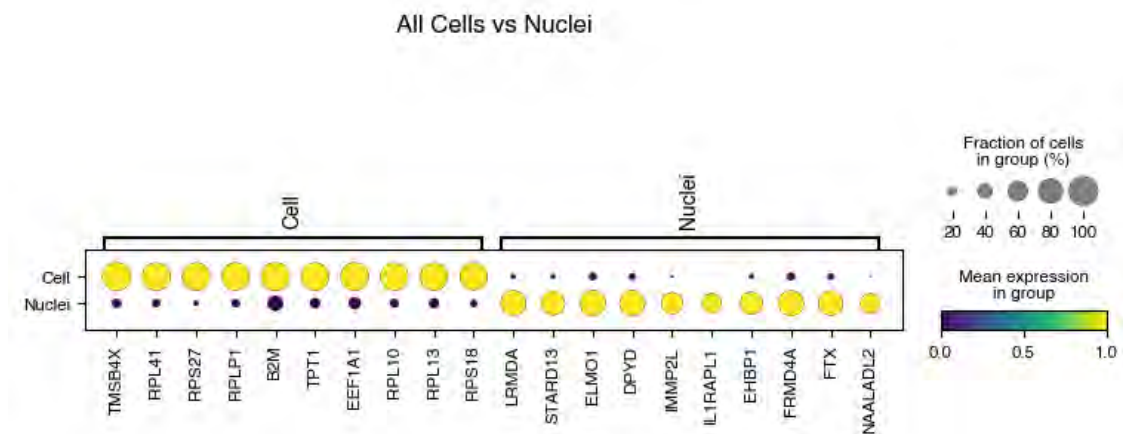
1 extra bytes in post.stringData array

'created' timestamp seems very low; regarding as unix timestamp

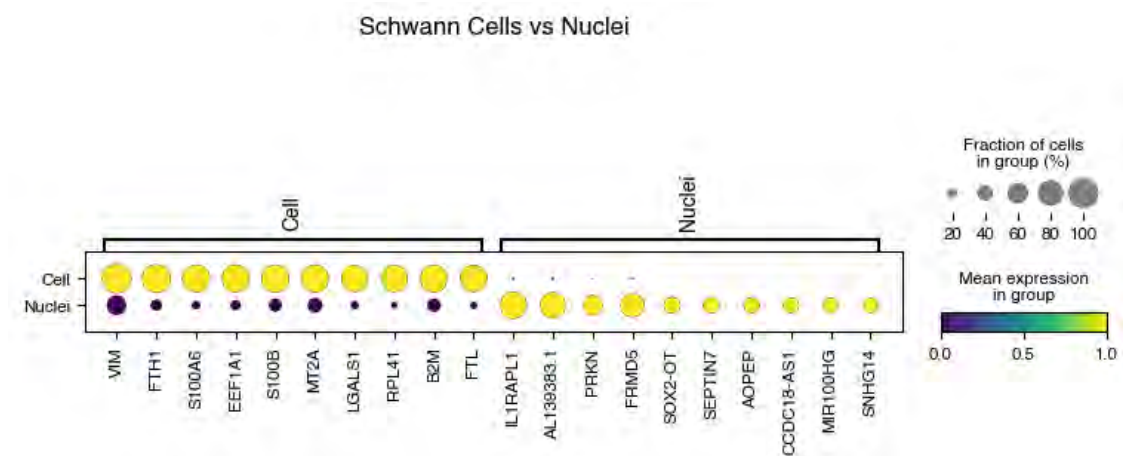
```
[52]: sc.pl.rank_genes_groups_dotplot(integrated, groupby='cell_vs_nuc', n_genes=10,
    ↪ use_raw=False, standard_scale='var', cmap='viridis', title='All Cells vs_
    ↪ Nuclei')
sc.pl.rank_genes_groups_dotplot(scwhann, groupby='cell_vs_nuc', n_genes=10,
    ↪ use_raw=False, standard_scale='var', cmap='viridis', title='Schwann Cells vs_
    ↪ Nuclei')
```

```
sc.pl.rank_genes_groups_dotplot(macro, groupby='cell_vs_nuc', n_genes=10,
    ↪use_raw=False, standard_scale='var', cmap='viridis', title='Macrophage Cells_
    ↪vs Nuclei')
sc.pl.rank_genes_groups_dotplot(endo, groupby='cell_vs_nuc', n_genes=10,
    ↪use_raw=False, standard_scale='var', cmap='viridis', title='Endothelial_
    ↪Cells vs Nuclei')
```

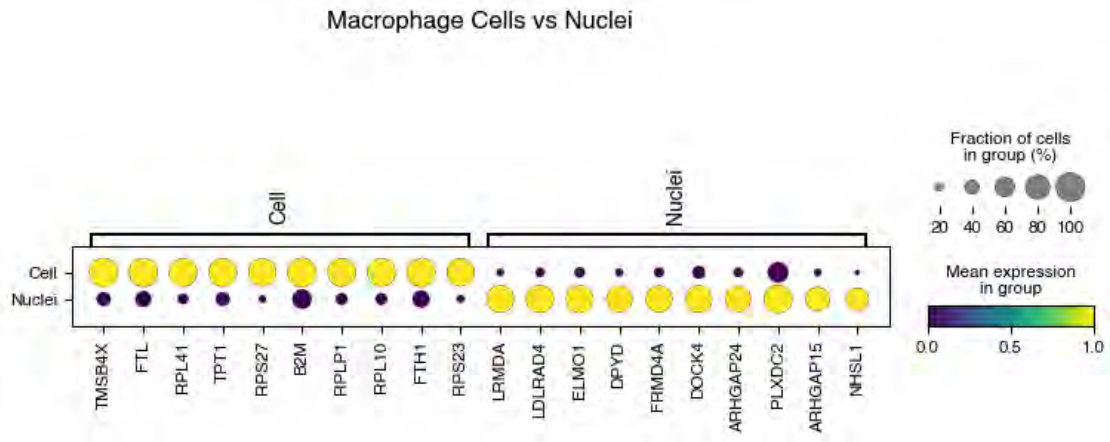
WARNING: Dendrogram not added. Dendrogram is added only when the number of categories to plot > 2



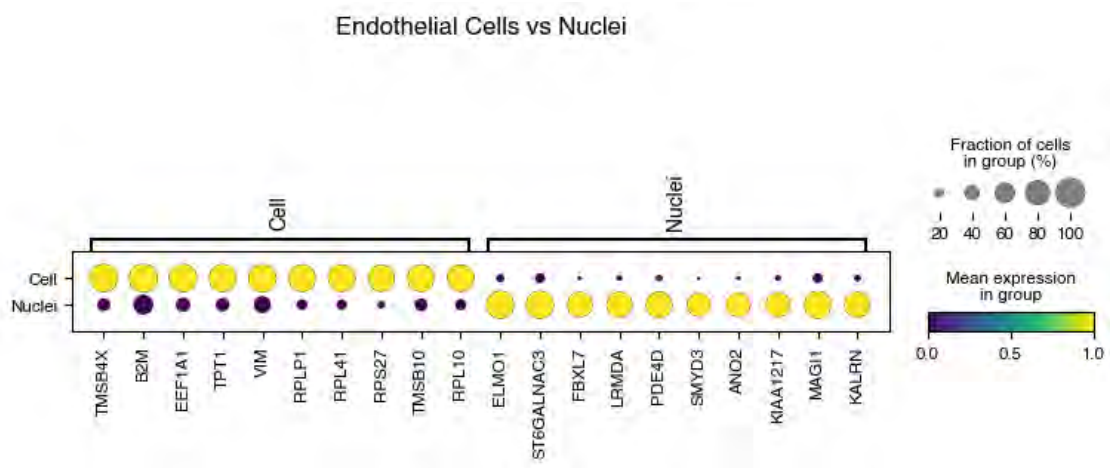
WARNING: Dendrogram not added. Dendrogram is added only when the number of categories to plot > 2



WARNING: Dendrogram not added. Dendrogram is added only when the number of categories to plot > 2



WARNING: Dendrogram not added. Dendrogram is added only when the number of categories to plot > 2



6.1 Figure 1D

```
[13]: ## Plot overall Integrated object colored by cell type
fig, ax1 = plt.subplots(nrows=1,ncols=1,figsize=(6,6), dpi=600)
ax1_dict = sc.pl.umap(integrated,
                        use_raw = False,
                        frameon=False,
                        show=False,
                        ax = ax1,
                        color = "pred.type",
                        s = 25,
                        palette = ct_dict,
```

```

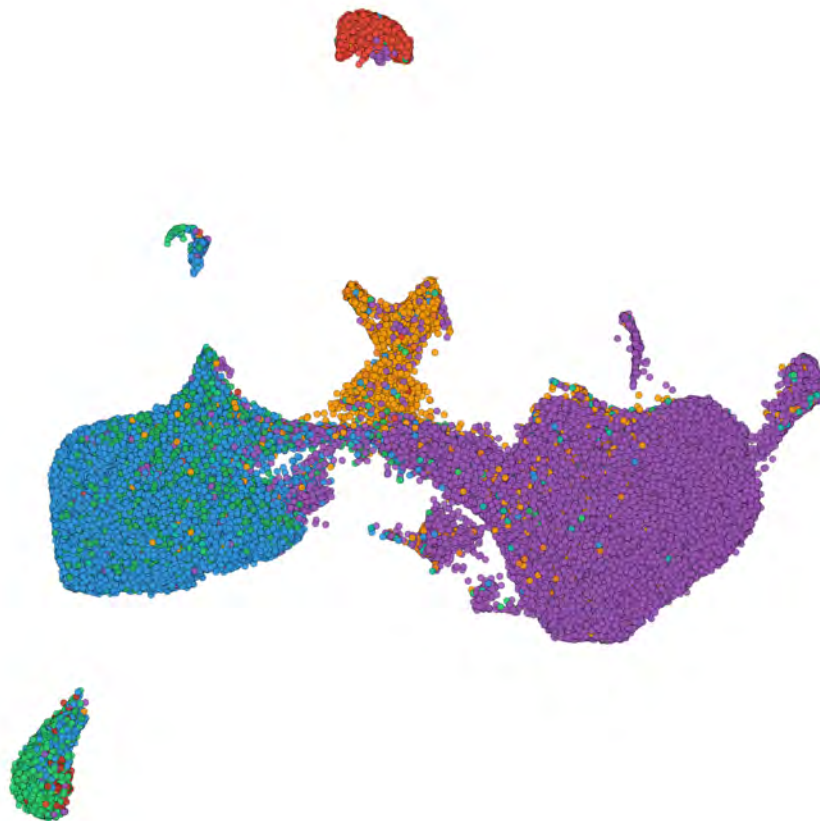
        edgecolor = 'black',
        linewidth= 0.1,
        alpha = 0.9,
        legend_loc=None,
        #legend_fontoutline=2
    )

#handles, labels = plt.gca().get_legend_handles_labels()
#
#leg1 = ax1.legend(prop = {'size':'14'}, markerscale=1.6, frameon=False,
#    ↪bbox_to_anchor=(1, 1))
#for lh in leg1.legend_handles:
#    lh.set_alpha(0.9)
#    lh.set_edgecolor("black")

ax1.set_title("")

fig.savefig('figure 1/cell_classes_01.pdf',bbox_inches='tight')
fig.savefig('figure 1/cell_classes_01.png',bbox_inches='tight')

```



7 Figure 2 NF2 haploinsufficiency

7.0.1 NF2 vs sporadic

```
[27]: ## Plot overall Integrated object colored by NF2 vs Sporadic
fig, ax1 = plt.subplots(nrows=1,ncols=1,figsize=(5,5), dpi=600)
ax1_dict = sc.pl.umap(integrated,
                      use_raw = False,
                      frameon=False,
                      show=False,
                      ax = ax1,
                      color = "type",
                      s = 25,
                      palette = NF2_spor_dict,
                      edgecolor = 'black',
                      linewidth= 0.2,
                      alpha = 0.9,
                      #legend_loc=None
                      )

handles, labels = plt.gca().get_legend_handles_labels()

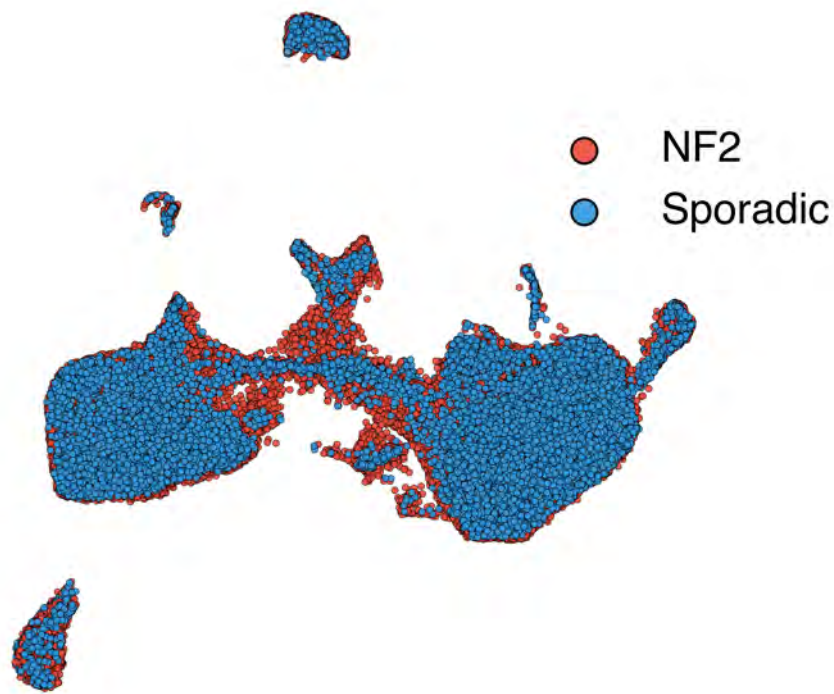
leg1 = ax1.legend(prop = {'size':'16', 'weight':'bold'}, markerscale=1.6,
                 ↪frameon=False, bbox_to_anchor=(0.7,0.85))
for lh in leg1.legend_handles:
    lh.set_alpha(0.9)
    lh.set_edgecolor("black")

ax1.set_title("")

fig.savefig('figure 1/int_NF2_spor.pdf',bbox_inches='tight')
fig.savefig('figure 1/int_NF2_spor.png',bbox_inches='tight')
```

1 extra bytes in post.stringData array

'created' timestamp seems very low; regarding as unix timestamp



```
[16]: nf2 = integrated[(integrated.obs['type']=='NF2') & (integrated.obs['x22q'].
    ↪isin(['22q+/+', '22q+/-', '22q-/-']))]
    spor = integrated[(integrated.obs['type']=='Sporadic') & (integrated.
    ↪obs['x22q'].isin(['22q+/+', '22q+/-', '22q-/-']))]
```

```
[87]: cell_dict = {'22q-/-': '#1f77b4',
    "22q+/-": '#ff7f0e',
    '22q+/+': '#2ca02c'}
fig, (ax1, ax2) = plt.subplots(nrows=1, ncols=2, figsize=(10, 5), dpi=600)
ax1_dict = sc.pl.umap(nf2,
    use_raw = False,
    frameon=False,
    show=False,
    ax = ax1,
    color = "x22q",
    s = 25,
    palette = cell_dict,
    edgecolor = 'black',
    linewidth= 0.4,
    alpha = 0.9
)
```

```

ax2_dict = sc.pl.umap(spor,
                      use_raw = False,
                      frameon=False,
                      show=False,
                      ax = ax2,
                      color = "x22q",
                      s = 25,
                      palette = cell_dict,
                      edgecolor = 'black',
                      linewidth= 0.4,
                      alpha = 0.9
                      )

handles, labels = plt.gca().get_legend_handles_labels()

leg1 = ax1.legend(prop = {'size':'14'}, markerscale=1.6, frameon=False,
                 bbox_to_anchor=(0, 1))
for lh in leg1.legend_handles:
    lh.set_alpha(0.9)
    lh.set_edgecolor("black")
leg2 = ax2.legend(prop = {'size':'14'}, markerscale=1.6, frameon=False,
                 bbox_to_anchor=(0, 1))
for lh in leg2.legend_handles:
    lh.set_alpha(0.9)
    lh.set_edgecolor("black")
ax1.set_title("")
ax2.set_title("")

fig.savefig('figure 1/NF2_Spor_22q.pdf',bbox_inches='tight')
fig.savefig('figure 1/NF2_Spor_22q.png',bbox_inches='tight')
fig.show()

```

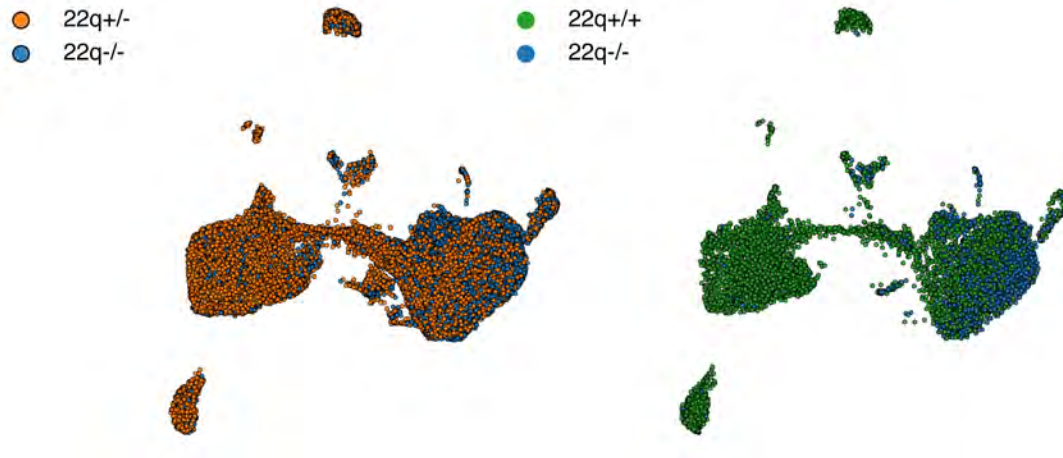
1 extra bytes in post.stringData array

'created' timestamp seems very low; regarding as unix timestamp

/var/folders/cq/kfh8gwnx4133mcyw_19df8j5ky9t11/T/ipykernel_44353/510990682.py:48

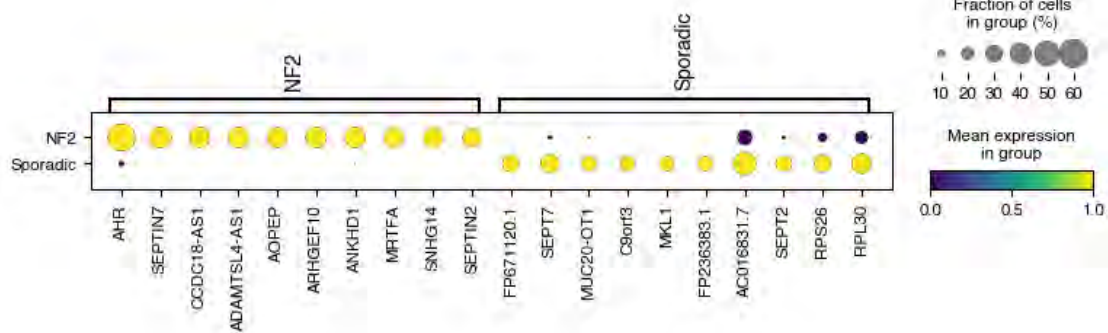
: UserWarning: FigureCanvasAgg is non-interactive, and thus cannot be shown

fig.show()

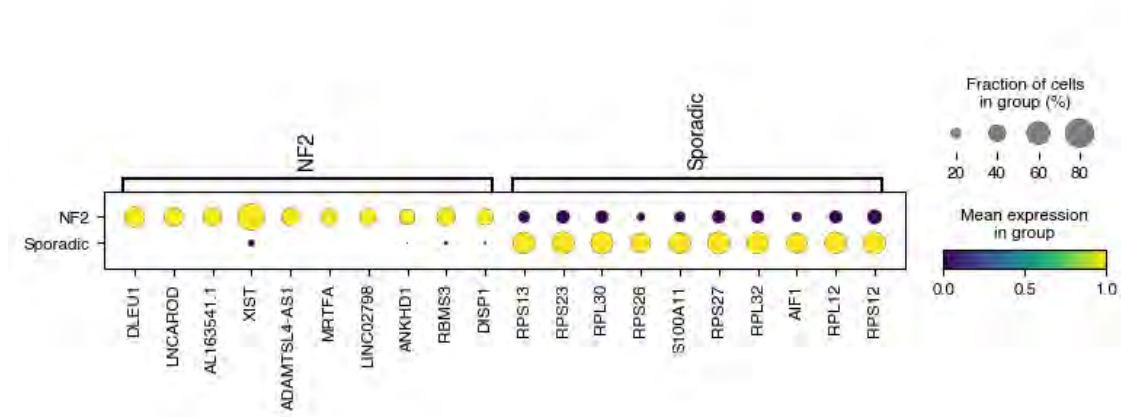


```
[30]: sc.pl.rank_genes_groups_dotplot(integrated, groupby='type', n_genes=10,
    ↪ use_raw=False, standard_scale='var', cmap='viridis')
sc.pl.rank_genes_groups_dotplot(macro, groupby='type', n_genes=10,
    ↪ use_raw=False, standard_scale='var', cmap='viridis')
sc.pl.rank_genes_groups_dotplot(schwann, groupby='type', n_genes=10,
    ↪ use_raw=False, standard_scale='var', cmap='viridis')
```

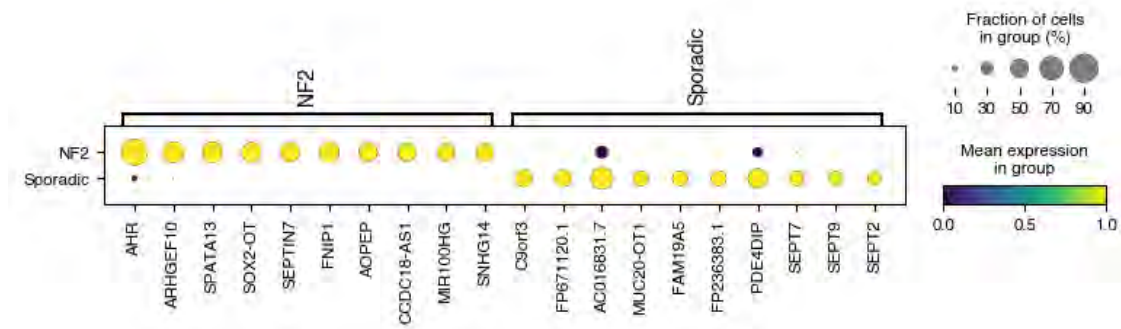
WARNING: Dendrogram not added. Dendrogram is added only when the number of categories to plot > 2



WARNING: Dendrogram not added. Dendrogram is added only when the number of categories to plot > 2



WARNING: Dendrogram not added. Dendrogram is added only when the number of categories to plot > 2



```
[ ]: sc.tl.rank_genes_groups(integrated, groupby='type', cell_types
```

7.1 Figure 2A All cells split by type

```
[14]: # subset nf2 and sporadic
# Subset single samples for plotting
int_nf2 = integrated[(integrated.obs['type'].isin(['NF2']))]
int_sporadic = integrated[(integrated.obs['type'].isin(['Sporadic']))]

[31]: ## Plot overall Integrated object colored by NF2 vs Sporadic
fig, (ax1, ax2) = plt.subplots(nrows=1,ncols=2,figsize=(10,5), dpi=600, )
ax1_grey = sc.pl.umap(integrated,
                        use_raw = False,
                        frameon=False,
```

```

        show=False,
        ax = ax1,
        # color = "pred.type",
        s = 30,
        # palette = ct_dict,
        # edgecolor = 'black',
        # linewidth= 0.5,
        alpha = 0.9,
    )
ax1_dict = sc.pl.umap(int_nf2,
                      use_raw = False,
                      frameon=False,
                      show=False,
                      ax = ax1,
                      color = "type",
                      s = 25,
                      palette = NF2_spor_dict,
                      edgecolor = 'black',
                      linewidth= 0.4,
                      alpha = 0.9,
                      )
ax2_grey = sc.pl.umap(integrated,
                      use_raw = False,
                      frameon=False,
                      show=False,
                      ax = ax2,
                      # color = "pred.type",
                      s = 30,
                      # palette = ct_dict,
                      # edgecolor = 'black',
                      # linewidth= 0.5,
                      alpha = 0.9,
                      )
ax1_dict = sc.pl.umap(int_sporadic,
                      use_raw = False,
                      frameon=False,
                      show=False,
                      ax = ax2,
                      color = "type",
                      s = 25,
                      palette = NF2_spor_dict,
                      edgecolor = 'black',
                      linewidth= 0.4,
                      alpha = 0.9,
                      )

handles, labels = plt.gca().get_legend_handles_labels()

```

```

leg1 = ax1.legend(prop = {'size':'14'}, markerscale=1.6, frameon=False,
↳bbox_to_anchor=(0, 1))
for lh in leg1.legend_handles:
    lh.set_alpha(0.9)
    lh.set_edgecolor("black")

leg2 = ax2.legend(prop = {'size':'14'}, markerscale=1.6, frameon=False,
↳bbox_to_anchor=(1, 1))
for lh in leg2.legend_handles:
    lh.set_alpha(0.9)
    lh.set_edgecolor("black")

ax1.set_title("")
ax2.set_title("")

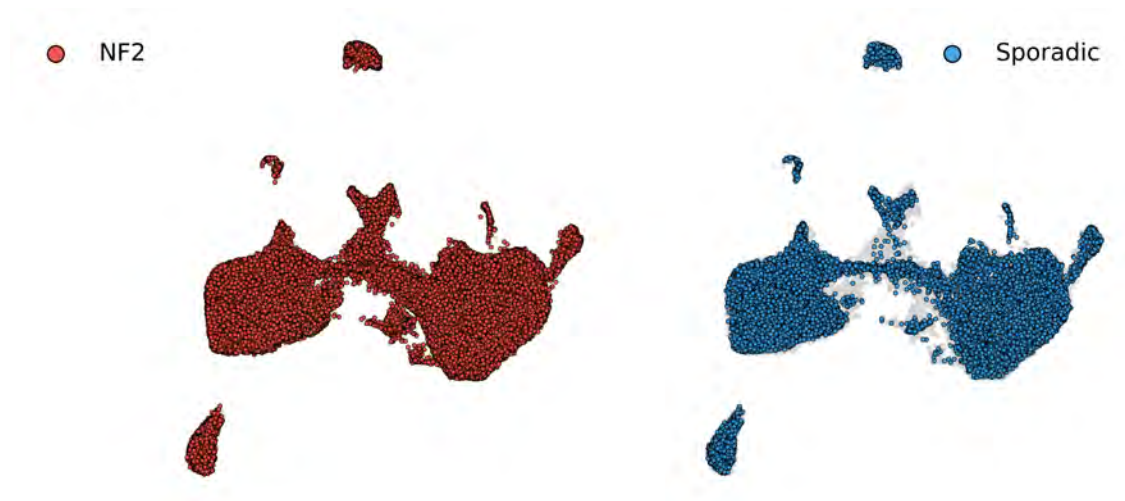
fig.savefig('figure 2/int_NF2_spor_double.pdf',bbox_inches='tight')
fig.savefig('figure 2/int_NF2_spor_double.png',bbox_inches='tight')

```

```

/Users/chittiboinap/anaconda3/lib/python3.11/site-
packages/scanpy/plotting/_tools/scatterplots.py:394: UserWarning: No data for
colormapping provided via 'c'. Parameters 'cmap', 'norm' will be ignored
    cax = scatter(
/Users/chittiboinap/anaconda3/lib/python3.11/site-
packages/scanpy/plotting/_tools/scatterplots.py:394: UserWarning: No data for
colormapping provided via 'c'. Parameters 'cmap' will be ignored
    cax = scatter(
/Users/chittiboinap/anaconda3/lib/python3.11/site-
packages/scanpy/plotting/_tools/scatterplots.py:394: UserWarning: No data for
colormapping provided via 'c'. Parameters 'cmap', 'norm' will be ignored
    cax = scatter(
/Users/chittiboinap/anaconda3/lib/python3.11/site-
packages/scanpy/plotting/_tools/scatterplots.py:394: UserWarning: No data for
colormapping provided via 'c'. Parameters 'cmap' will be ignored
    cax = scatter(

```



7.2 Figure 2B Schwann cells by subclasses and Type

```
[12]: sc.pl.umap(schwann,
                use_raw = False,
                frameon=False,
                show=False,
                #ax = ax1,
                color = "schwann_subclasses",
                s = 25,
                palette = schwann_dict,
                edgecolor = 'black',
                linewidth= 0.4,
                alpha = 0.9,
                vmin=1,
                # vmax=10,
                legend_loc='on data'
                )
```

```
-----
NameError                                Traceback (most recent call last)
Cell In[12], line 1
----> 1 sc.pl.umap(schwann,
      2
      3         use_raw = False,
      4         frameon=False,
      5         show=False,
      6         #ax = ax1,
      7         color = "schwann_subclasses",
      8         s = 25,
      9         palette = schwann_dict,
      10        edgecolor = 'black',
```

```

10             linewidth= 0.4,
11             alpha = 0.9,
12             vmin=1,
13             # vmax=10,
14             legend_loc='on data'
15         )

```

NameError: name 'schwann' is not defined

```

[1]: ## Plot schwann subset
fig, (ax1, ax2) = plt.subplots(nrows=1,ncols=2,figsize=(12,6), dpi=600)
ax1_dict = sc.pl.umap(schwann,
                      use_raw = False,
                      frameon=False,
                      show=False,
                      ax = ax1,
                      color = "schwann_subclasses",
                      s = 25,
                      palette = schwann_dict,
                      edgecolor = 'black',
                      linewidth= 0.4,
                      alpha = 0.9,
                      vmin=1,
                      # vmax=5
                      legen
                      legend_loc='on data'
                      )
ax2_dict = sc.pl.umap(schwann,
                      use_raw = False,
                      frameon=False,
                      show=False,
                      ax = ax2,
                      color = "type",
                      s = 25,
                      palette = NF2_spor_dict,
                      edgecolor = 'black',
                      linewidth= 0.4,
                      alpha = 0.9,
                      )
leg1 = ax1.legend(prop = {'size':'14'}, markerscale=1.6, frameon=False,
                 ↪ bbox_to_anchor=(0, 1))
for lh in leg1.legend_handles:
    lh.set_alpha(0.9)
    lh.set_edgecolor("black")

```

```

leg2 = ax2.legend(prop = {'size':'14'}, markerscale=1.6, frameon=False,
↳ bbox_to_anchor=(1, 1))
for lh in leg2.legend_handles:
    lh.set_alpha(0.9)
    lh.set_edgecolor("black")

ax1.set_title("Schwann Cell Subclasses", fontweight="bold", fontsize=24)
ax2.set_title("Germline status", fontweight="bold", fontsize=24)
#fig.savefig('figure 2/Schwann_cells.png',bbox_inches='tight')
#fig.savefig('figure 2/Schwann_cells.pdf',bbox_inches='tight')

```

Cell In[1], line 16

legen

^

SyntaxError: invalid syntax. Perhaps you forgot a comma?

7.3 Figure 2C Schwann by type

```

[3]: schwann = sc.read_h5ad("/Users/chittiboinap/Library/CloudStorage/
↳ OneDrive-NationalInstitutesofHealth/NIH Research/NF2/nf2 schwannoma paper/
↳ data/Ihika_Analysis/results/integrated/integrated_new/Schwann_GEX.h5ad")

```

```

[4]: schwann

```

```

[4]: AnnData object with n_obs × n_vars = 49639 × 38224
      obs: 'nCount_RNA', 'nFeature_RNA', 'orig.ident', 'type', 'percent.mt',
'pANN_0.25_0.09_0', 'DF.classifications_0.25_0.09_0', 'RNA_snn_res.0.5',
'seurat_clusters', 'pred.type', 'score1', 'X1p', 'X1q', 'X2p', 'X2q', 'X3p',
'X3q', 'X4p', 'X4q', 'X5p', 'X5q', 'X6p', 'X6q', 'X7p', 'X7q', 'X8p', 'X8q',
'X9p', 'X9q', 'X10p', 'X10q', 'X11p', 'X11q', 'X12p', 'X12q', 'X13q', 'X14q',
'X15q', 'X16p', 'X16q', 'X17p', 'X17q', 'X18q', 'X19p', 'X19q', 'X20p', 'X20q',
'X21q', 'X22q', 'x22q', 'integrated_snn_res.0.5', 'integrated_snn_res.1',
'schwann_pred.type', 'integrated_snn_res.0.3', 'S.Score', 'G2M.Score', 'Phase',
'old.ident', 'integrated_snn_res.0.1', 'schwann_subclasses', 'score21', 'Hippo1'
      var: 'features'
      obsm: 'X_umap'

```

```

[35]: # subset Schwann cells nf2 and sporadic
# Subset single samples for plotting
sch_nf2 = schwann[(schwann.obs['type'].isin(['NF2']))]
sch_sporadic = schwann[(schwann.obs['type'].isin(['Sporadic']))]

```

```

[37]: ## Plot schwann subset
fig, (ax1,ax2) = plt.subplots(nrows=1,ncols=2,figsize=(9,4), dpi=600)
ax1_grey = sc.pl.umap(schwann,

```

```

        use_raw = False,
        frameon=False,
        show=False,
        ax = ax1,
        # color = "pred.type",
        s = 30,
        # palette = ct_dict,
        # edgecolor = 'black',
        # linewidth= 0.5,
        alpha = 0.9,
    )
ax1_dict = sc.pl.umap(sch_nf2,
    use_raw = False,
    frameon=False,
    show=False,
    ax = ax1,
    color = "type",
    s = 25,
    palette = NF2_spor_dict,
    edgecolor = 'black',
    linewidth= 0.4,
    alpha = 0.9,
)
ax2_grey = sc.pl.umap(schwann,
    use_raw = False,
    frameon=False,
    show=False,
    ax = ax2,
    # color = "pred.type",
    s = 30,
    # palette = ct_dict,
    # edgecolor = 'black',
    # linewidth= 0.5,
    alpha = 0.9,
)
ax1_dict = sc.pl.umap(sch_sporadic,
    use_raw = False,
    frameon=False,
    show=False,
    ax = ax2,
    color = "type",
    s = 25,
    palette = NF2_spor_dict,
    edgecolor = 'black',
    linewidth= 0.4,
    alpha = 0.9,
)

```

```

handles, labels = plt.gca().get_legend_handles_labels()
ax1.set_title("")
ax2.set_title("")

leg1 = ax1.legend(prop = {'size':'14'}, markerscale=1.6, frameon=False,
    ↳ bbox_to_anchor=(0.1, 1))
for lh in leg1.legend_handles:
    lh.set_alpha(0.9)
    lh.set_edgecolor("black")

leg2 = ax2.legend(prop = {'size':'14'}, markerscale=1.6, frameon=False,
    ↳ bbox_to_anchor=(0.9, 1))
for lh in leg2.legend_handles:
    lh.set_alpha(0.9)
    lh.set_edgecolor("black")

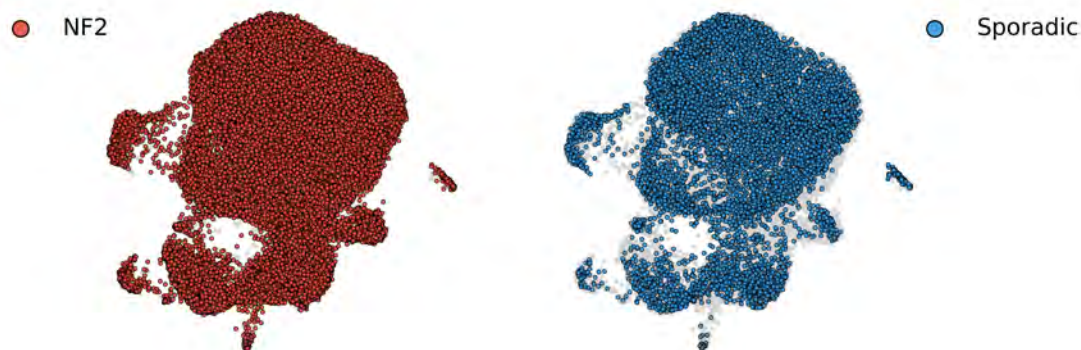
fig.savefig('figure 2/Schwann_cell_type.png', bbox_inches='tight')
fig.savefig('figure 2/Schwann_cell_type.pdf', bbox_inches='tight')

```

```

/Users/chittiboinap/anaconda3/lib/python3.11/site-
packages/scanpy/plotting/_tools/scatterplots.py:394: UserWarning: No data for
colormapping provided via 'c'. Parameters 'cmap', 'norm' will be ignored
    cax = scatter(
/Users/chittiboinap/anaconda3/lib/python3.11/site-
packages/scanpy/plotting/_tools/scatterplots.py:394: UserWarning: No data for
colormapping provided via 'c'. Parameters 'cmap' will be ignored
    cax = scatter(
/Users/chittiboinap/anaconda3/lib/python3.11/site-
packages/scanpy/plotting/_tools/scatterplots.py:394: UserWarning: No data for
colormapping provided via 'c'. Parameters 'cmap', 'norm' will be ignored
    cax = scatter(
/Users/chittiboinap/anaconda3/lib/python3.11/site-
packages/scanpy/plotting/_tools/scatterplots.py:394: UserWarning: No data for
colormapping provided via 'c'. Parameters 'cmap' will be ignored
    cax = scatter(

```



8 Macrophage Subclassifications NF2 vs Sporadic

```
[73]: macro = sc.read_h5ad("/Users/chittiboinap/Library/CloudStorage/  
↳OneDrive-NationalInstitutesofHealth/NIH Research/NF2/nf2 schwannoma paper/  
↳data/Ihika_Analysis/results/integrated/integrated_new/Macrophage_GEX.h5ad")
```

```
/Users/chittiboinap/anaconda3/envs/spatial/lib/python3.9/site-  
packages/anndata/compat/__init__.py:311: FutureWarning: Moving element from  
.uns['neighbors']['distances'] to .obsp['distances'].
```

This is where adjacency matrices should go now.
warn(

```
[74]: macro
```

```
[74]: AnnData object with n_obs × n_vars = 29776 × 2000  
      obs: 'nCount_RNA', 'nFeature_RNA', 'orig.ident', 'type', 'percent.mt',  
      'pANN_0.25_0.09_0', 'DF.classifications_0.25_0.09_0', 'RNA_snn_res.0.5',  
      'seurat_clusters', 'pred.type', 'score1', 'X1p', 'X1q', 'X2p', 'X2q', 'X3p',  
      'X3q', 'X4p', 'X4q', 'X5p', 'X5q', 'X6p', 'X6q', 'X7p', 'X7q', 'X8p', 'X8q',  
      'X9p', 'X9q', 'X10p', 'X10q', 'X11p', 'X11q', 'X12p', 'X12q', 'X13q', 'X14q',  
      'X15q', 'X16p', 'X16q', 'X17p', 'X17q', 'X18q', 'X19p', 'X19q', 'X20p', 'X20q',  
      'X21q', 'X22q', 'x22q', 'integrated_snn_res.0.5', 'integrated_snn_res.0.1',  
      'integrated_snn_res.0.3', 'macrophage_subclasses', 'Kim_myeloid'  
      var: 'vst.mean', 'vst.variance', 'vst.variance.expected',  
      'vst.variance.standardized', 'vst.variable'  
      uns: 'neighbors'  
      obsm: 'X_pca', 'X_umap'  
      varm: 'PCs'  
      obsp: 'distances'
```

```
[85]: # subset Schwann cells nf2 and sporadic  
      # Subset single samples for plotting  
      macro_nf2 = macro[(macro.obs['type'].isin(['NF2']))]  
      macro_sporadic = macro[(macro.obs['type'].isin(['Sporadic']))]
```

8.0.1 Plot macrophages by 'type' and subclasses

```
[93]: ## Plot macrophages by 'type'  
      fig, (ax1, ax2, ax3) = plt.subplots(nrows=1, ncols=3, figsize=(12,4), dpi=600)  
      ax1_grey = sc.pl.umap(macro,  
                           use_raw = False,  
                           frameon=False,  
                           show=False,
```

```

        ax = ax1,
        # color = "pred.type",
        s = 30,
        # palette = ct_dict,
        # edgecolor = 'black',
        # linewidth= 0.5,
        alpha = 0.9,
    )
ax1_dict = sc.pl.umap(macro_nf2,
    use_raw = False,
    frameon=False,
    show=False,
    ax = ax1,
    color = "type",
    s = 25,
    #palette = macro_dict1,
    edgecolor = 'black',
    linewidth= 0.4,
    alpha = 0.9,
)
ax2_grey = sc.pl.umap(macro,
    use_raw = False,
    frameon=False,
    show=False,
    ax = ax2,
    # color = "pred.type",
    s = 30,
    # palette = ct_dict,
    # edgecolor = 'black',
    # linewidth= 0.5,
    alpha = 0.9,
)
ax2_dict = sc.pl.umap(macro_sporadic,
    use_raw = False,
    frameon=False,
    show=False,
    ax = ax2,
    color = "type",
    s = 25,
    #palette = macro_dict1,
    edgecolor = 'black',
    linewidth= 0.4,
    alpha = 0.9,
)
ax3_grey = sc.pl.umap(macro,
    use_raw = False,
    frameon=False,

```

```

        show=False,
        ax = ax3,
        # color = "pred.type",
        s = 30,
        # palette = ct_dict,
        # edgecolor = 'black',
        # linewidth= 0.5,
        alpha = 0.9,
    )
ax3_dict = sc.pl.umap(macro,
                      use_raw = False,
                      frameon=False,
                      show=False,
                      ax = ax3,
                      color = "macrophage_subclasses",
                      s = 25,
                      palette = macro_dict1,
                      edgecolor = 'black',
                      linewidth= 0.4,
                      alpha = 0.9,
                      )
handles, labels = plt.gca().get_legend_handles_labels()
ax1.set_title("")
ax2.set_title("")

leg1 = ax1.legend(prop = {'size':'14'}, markerscale=1.6, frameon=False,
                 ↪bbox_to_anchor=(0.1, 1))
for lh in leg1.legend_handles:
    lh.set_alpha(0.9)
    lh.set_edgecolor("black")

leg2 = ax2.legend(prop = {'size':'14'}, markerscale=1.6, frameon=False,
                 ↪bbox_to_anchor=(0.9, 1))
for lh in leg2.legend_handles:
    lh.set_alpha(0.9)
    lh.set_edgecolor("black")

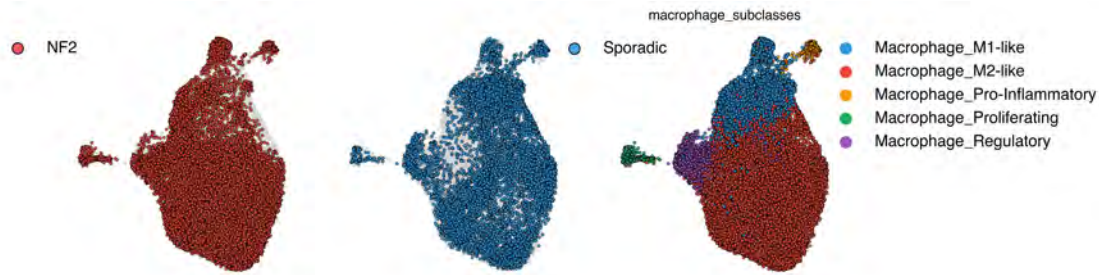
leg3 = ax3.legend(prop = {'size':'14'}, markerscale=1.6, frameon=False,
                 ↪bbox_to_anchor=(0.9, 1))
for lh in leg3.legend_handles:
    lh.set_alpha(0.9)
    lh.set_edgecolor("black")
fig.savefig('supplemental figure x/macrophage_distribution.
           ↪png',bbox_inches='tight')
fig.savefig('supplemental figure x/macrophage_distribution.
           ↪pdf',bbox_inches='tight')

```

```

/Users/chittiboinap/anaconda3/envs/spatial/lib/python3.9/site-
packages/scanpy/plotting/_tools/scatterplots.py:394: UserWarning: No data for
colormapping provided via 'c'. Parameters 'cmap', 'norm' will be ignored
    cax = scatter(
/Users/chittiboinap/anaconda3/envs/spatial/lib/python3.9/site-
packages/scanpy/plotting/_tools/scatterplots.py:1234: FutureWarning: The default
value of 'ignore' for the `na_action` parameter in pandas.Categorical.map is
deprecated and will be changed to 'None' in a future version. Please set
na_action to the desired value to avoid seeing this warning
    color_vector = pd.Categorical(values.map(color_map))
/Users/chittiboinap/anaconda3/envs/spatial/lib/python3.9/site-
packages/scanpy/plotting/_tools/scatterplots.py:394: UserWarning: No data for
colormapping provided via 'c'. Parameters 'cmap' will be ignored
    cax = scatter(
/Users/chittiboinap/anaconda3/envs/spatial/lib/python3.9/site-
packages/scanpy/plotting/_tools/scatterplots.py:394: UserWarning: No data for
colormapping provided via 'c'. Parameters 'cmap', 'norm' will be ignored
    cax = scatter(
/Users/chittiboinap/anaconda3/envs/spatial/lib/python3.9/site-
packages/scanpy/plotting/_tools/scatterplots.py:1234: FutureWarning: The default
value of 'ignore' for the `na_action` parameter in pandas.Categorical.map is
deprecated and will be changed to 'None' in a future version. Please set
na_action to the desired value to avoid seeing this warning
    color_vector = pd.Categorical(values.map(color_map))
/Users/chittiboinap/anaconda3/envs/spatial/lib/python3.9/site-
packages/scanpy/plotting/_tools/scatterplots.py:394: UserWarning: No data for
colormapping provided via 'c'. Parameters 'cmap' will be ignored
    cax = scatter(
/Users/chittiboinap/anaconda3/envs/spatial/lib/python3.9/site-
packages/scanpy/plotting/_tools/scatterplots.py:394: UserWarning: No data for
colormapping provided via 'c'. Parameters 'cmap', 'norm' will be ignored
    cax = scatter(
/Users/chittiboinap/anaconda3/envs/spatial/lib/python3.9/site-
packages/scanpy/plotting/_tools/scatterplots.py:1234: FutureWarning: The default
value of 'ignore' for the `na_action` parameter in pandas.Categorical.map is
deprecated and will be changed to 'None' in a future version. Please set
na_action to the desired value to avoid seeing this warning
    color_vector = pd.Categorical(values.map(color_map))
/Users/chittiboinap/anaconda3/envs/spatial/lib/python3.9/site-
packages/scanpy/plotting/_tools/scatterplots.py:394: UserWarning: No data for
colormapping provided via 'c'. Parameters 'cmap' will be ignored
    cax = scatter(
1 extra bytes in post.stringData array
'created' timestamp seems very low; regarding as unix timestamp
Zapf NOT subset; don't know how to subset; dropped
feat NOT subset; don't know how to subset; dropped
meta NOT subset; don't know how to subset; dropped
morx NOT subset; don't know how to subset; dropped

```



8.1 Create DEGs between NF2 and Sporadic macrophages

```
[75]: sc.tl.rank_genes_groups(macro, groupby="type")
```

WARNING: Default of the method has been changed to 't-test' from 't-test_overestim_var'

/Users/chittiboinap/anaconda3/envs/spatial/lib/python3.9/site-packages/scanpy/tools/_rank_genes_groups.py:422: RuntimeWarning: invalid value encountered in log2

self.stats[group_name, 'logfoldchanges'] = np.log2(
/Users/chittiboinap/anaconda3/envs/spatial/lib/python3.9/site-packages/scanpy/tools/_rank_genes_groups.py:422: RuntimeWarning: invalid value encountered in log2

self.stats[group_name, 'logfoldchanges'] = np.log2(

```
[76]: macro
```

```
[76]: AnnData object with n_obs × n_vars = 29776 × 2000
```

```
obs: 'nCount_RNA', 'nFeature_RNA', 'orig.ident', 'type', 'percent.mt',  
'pANN_0.25_0.09_0', 'DF.classifications_0.25_0.09_0', 'RNA_snn_res.0.5',  
'seurat_clusters', 'pred.type', 'score1', 'X1p', 'X1q', 'X2p', 'X2q', 'X3p',  
'X3q', 'X4p', 'X4q', 'X5p', 'X5q', 'X6p', 'X6q', 'X7p', 'X7q', 'X8p', 'X8q',  
'X9p', 'X9q', 'X10p', 'X10q', 'X11p', 'X11q', 'X12p', 'X12q', 'X13q', 'X14q',  
'X15q', 'X16p', 'X16q', 'X17p', 'X17q', 'X18q', 'X19p', 'X19q', 'X20p', 'X20q',  
'X21q', 'X22q', 'x22q', 'integrated_snn_res.0.5', 'integrated_snn_res.0.1',  
'integrated_snn_res.0.3', 'macrophage_subclasses', 'Kim_myeloid'  
var: 'vst.mean', 'vst.variance', 'vst.variance.expected',  
'vst.variance.standardized', 'vst.variable'  
uns: 'neighbors', 'rank_genes_groups'  
obsm: 'X_pca', 'X_umap'  
varm: 'PCs'  
obsp: 'distances'
```

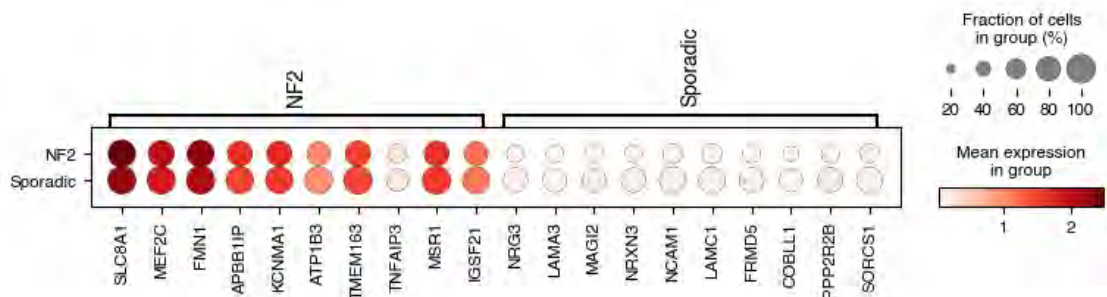
```
[77]: sc.pl.rank_genes_groups_dotplot(macro)
```

WARNING: Dendrogram not added. Dendrogram is added only when the number of categories to plot > 2

```

/Users/chittiboinap/anaconda3/envs/spatial/lib/python3.9/site-
packages/scanpy/get/get.py:69: FutureWarning: The previous implementation of
stack is deprecated and will be removed in a future version of pandas. See the
What's New notes for pandas 2.1.0 for details. Specify future_stack=True to
adopt the new implementation and silence this warning.
    d = d.stack(level=1).reset_index()
/Users/chittiboinap/anaconda3/envs/spatial/lib/python3.9/site-
packages/scanpy/get/get.py:69: FutureWarning: The previous implementation of
stack is deprecated and will be removed in a future version of pandas. See the
What's New notes for pandas 2.1.0 for details. Specify future_stack=True to
adopt the new implementation and silence this warning.
    d = d.stack(level=1).reset_index()
/Users/chittiboinap/anaconda3/envs/spatial/lib/python3.9/site-
packages/scanpy/plotting/_dotplot.py:168: FutureWarning: The default of
observed=False is deprecated and will be changed to True in a future version of
pandas. Pass observed=False to retain current behavior or observed=True to adopt
the future default and silence this warning.
    obs_bool.groupby(level=0).sum() / obs_bool.groupby(level=0).count()
/Users/chittiboinap/anaconda3/envs/spatial/lib/python3.9/site-
packages/scanpy/plotting/_dotplot.py:178: FutureWarning: The default of
observed=False is deprecated and will be changed to True in a future version of
pandas. Pass observed=False to retain current behavior or observed=True to adopt
the future default and silence this warning.
    dot_color_df = self.obs_tidy.groupby(level=0).mean()
/Users/chittiboinap/anaconda3/envs/spatial/lib/python3.9/site-
packages/scanpy/plotting/_dotplot.py:747: UserWarning: No data for colormapping
provided via 'c'. Parameters 'cmap', 'norm' will be ignored
    dot_ax.scatter(x, y, **kwargs)

```



8.1.1 GSEA for NF2 macrophages

```
[29]: df=pd.read_csv('supplemental figure x/GSEA/macro_type_DEG_NF2.csv', index_col=0)
      column_names_list = df.columns.tolist()
      print(column_names_list)
      print(df.head())
```

```
['names', 'scores', 'logfoldchanges', 'pvals', 'pvals_adj']
      names      scores  logfoldchanges      pvals      pvals_adj
0   SLC8A1  15.391752      0.333604  4.023965e-53  2.682643e-50
1   MEF2C  11.786323      0.285852  6.034611e-32  2.806796e-30
2    FMN1  11.598105      0.245030  5.276109e-31  2.110443e-29
3  APBB1IP   7.945061      0.203580  2.054294e-15  2.490053e-14
4   KCNMA1   7.552537      0.189810  4.471149e-14  4.940497e-13
```

```
df_up = df[(df['logfoldchanges'] > 1) & (df['pvals_adj'] < 0.05)] df_up
```

```
[30]: #df['Rank'] = -np.log10(df.pvals_adj)*df.logfoldchanges
      df = df.rename(columns = {'names':'Gene', 'scores':'Rank'})
      df = df.sort_values('Rank', ascending=False).reset_index(drop=True)
      df
```

```
[30]:      Gene      Rank  logfoldchanges      pvals      pvals_adj
0   SLC8A1  15.391752      0.333604  4.023965e-53  2.682643e-50
1   MEF2C  11.786323      0.285852  6.034611e-32  2.806796e-30
2    FMN1  11.598105      0.245030  5.276109e-31  2.110443e-29
3  APBB1IP   7.945061      0.203580  2.054294e-15  2.490053e-14
4   KCNMA1   7.552537      0.189810  4.471149e-14  4.940497e-13
...
1995  NCAM1 -14.537348     -0.679915  1.281036e-47  4.270118e-45
1996  NRXN3 -14.574862     -0.876969  7.745418e-48  3.098167e-45
1997  MAGI2 -15.183697     -1.009954  1.021515e-51  5.107575e-49
1998  LAMA3 -15.404770     -1.102958  3.603018e-53  2.682643e-50
1999   NRG3 -16.818535     -0.948152  4.545700e-63  9.091400e-60
```

```
[2000 rows x 5 columns]
```

```
[31]: ranking = df[['Gene', 'Rank']]
      ranking
```

```
[31]:      Gene      Rank
0   SLC8A1  15.391752
1   MEF2C  11.786323
2    FMN1  11.598105
3  APBB1IP   7.945061
4   KCNMA1   7.552537
...
1995  NCAM1 -14.537348
1996  NRXN3 -14.574862
```

```

1997    MAGI2 -15.183697
1998    LAMA3 -15.404770
1999    NRG3 -16.818535

```

```
[2000 rows x 2 columns]
```

```
[34]: gp.get_library_name()
```

```

[34]: ['ARCHS4_Cell-lines',
      'ARCHS4_IDG_Coexp',
      'ARCHS4_Kinases_Coexp',
      'ARCHS4_TFs_Coexp',
      'ARCHS4_Tissues',
      'Achilles_fitness_decrease',
      'Achilles_fitness_increase',
      'Aging_Perturbations_from_GEO_down',
      'Aging_Perturbations_from_GEO_up',
      'Allen_Brain_Atlas_10x_scRNA_2021',
      'Allen_Brain_Atlas_down',
      'Allen_Brain_Atlas_up',
      'Azimuth_2023',
      'Azimuth_Cell_Types_2021',
      'BioCarta_2013',
      'BioCarta_2015',
      'BioCarta_2016',
      'BioPlanet_2019',
      'BioPlex_2017',
      'CCLE_Proteomics_2020',
      'CM4AI_U2OS_Protein_Localization_Assemblies',
      'COMPARTMENTS_Curated_2025',
      'COMPARTMENTS_Experimental_2025',
      'CORUM',
      'COVID-19_Related_Gene_Sets',
      'COVID-19_Related_Gene_Sets_2021',
      'Cancer_Cell_Line_Encyclopedia',
      'CellMarker_2024',
      'CellMarker_Augmented_2021',
      'ChEA_2013',
      'ChEA_2015',
      'ChEA_2016',
      'ChEA_2022',
      'Chromosome_Location',
      'Chromosome_Location_hg19',
      'ClinVar_2019',
      'ClinVar_2025',
      'DGIdb_Drug_Targets_2024',
      'DSigDB',

```

'Data_Acquisition_Method_Most_Popular_Genes',
 'DepMap_CRISPR_GeneDependency_CellLines_2023',
 'DepMap_WG_CRISPR_Screens_Broad_CellLines_2019',
 'DepMap_WG_CRISPR_Screens_Sanger_CellLines_2019',
 'Descartes_Cell_Types_and_Tissue_2021',
 'Diabetes_Perturbations_GEO_2022',
 'DisGeNET',
 'Disease_Perturbations_from_GEO_down',
 'Disease_Perturbations_from_GEO_up',
 'Disease_Signatures_from_GEO_down_2014',
 'Disease_Signatures_from_GEO_up_2014',
 'DrugMatrix',
 'Drug_Perturbations_from_GEO_2014',
 'Drug_Perturbations_from_GEO_down',
 'Drug_Perturbations_from_GEO_up',
 'ENCODE_Histone_Modifications_2013',
 'ENCODE_Histone_Modifications_2015',
 'ENCODE_TF_ChIP-seq_2014',
 'ENCODE_TF_ChIP-seq_2015',
 'ENCODE_and_ChEA_Consensus_TFs_from_ChIP-X',
 'ESCAPE',
 'Elsevier_Pathway_Collection',
 'Enrichr_Libraries_Most_Popular_Genes',
 'Enrichr_Submissions_TF-Gene_Cooccurrence',
 'Enrichr_Users_Contributed_Lists_2020',
 'Epigenomics_Roadmap_HM_ChIP-seq',
 'FANTOM6_lncRNA_KD_DEGs',
 'GO_Biological_Process_2021',
 'GO_Biological_Process_2023',
 'GO_Biological_Process_2025',
 'GO_Cellular_Component_2021',
 'GO_Cellular_Component_2023',
 'GO_Cellular_Component_2025',
 'GO_Molecular_Function_2021',
 'GO_Molecular_Function_2023',
 'GO_Molecular_Function_2025',
 'GTEx_Aging_Signatures_2021',
 'GTEx_Tissue_Expression_Down',
 'GTEx_Tissue_Expression_Up',
 'GTEx_Tissues_V8_2023',
 'GWAS_Catalog_2019',
 'GWAS_Catalog_2023',
 'GWAS_Catalog_2025',
 'GeDiPNet_2023',
 'GeneSigDB',
 'Gene_Perturbations_from_GEO_down',
 'Gene_Perturbations_from_GEO_up',

'Genes_Associated_with_NIH_Grants',
 'Genome_Browser_PWMs',
 'GlyGen_Glycosylated_Proteins_2022',
 'HDSigDB_Human_2021',
 'HDSigDB_Mouse_2021',
 'HMDB_Metabolites',
 'HMS_LINCS_KinomeScan',
 'HomoloGene',
 'HuBMAP_ASCT_plus_B_augmented_w_RNAseq_Coexpression',
 'HuBMAP_ASCTplusB_augmented_2022',
 'HumanCyc_2015',
 'HumanCyc_2016',
 'Human_Gene_Atlas',
 'Human_Phenotype_Ontology',
 'IDG_Drug_Targets_2022',
 'InterPro_Domains_2019',
 'JASPAR_PWM_Human_2025',
 'JASPAR_PWM_Mouse_2025',
 'Jensen_COMPARTMENTS',
 'Jensen_DISEASES',
 'Jensen_DISEASES_Curated_2025',
 'Jensen_DISEASES_Experimental_2025',
 'Jensen_TISSUES',
 'KEA_2013',
 'KEA_2015',
 'KEGG_2013',
 'KEGG_2015',
 'KEGG_2016',
 'KEGG_2019_Human',
 'KEGG_2019_Mouse',
 'KEGG_2021_Human',
 'KOMP2_Mouse_Phenotypes_2022',
 'Kinase_Perturbations_from_GEO_down',
 'Kinase_Perturbations_from_GEO_up',
 'L1000_Kinase_and_GPCR_Perturbations_down',
 'L1000_Kinase_and_GPCR_Perturbations_up',
 'LINCS_L1000_CRISPR_KO_Consensus_Sigs',
 'LINCS_L1000_Chem_Pert_Consensus_Sigs',
 'LINCS_L1000_Chem_Pert_down',
 'LINCS_L1000_Chem_Pert_up',
 'LINCS_L1000_Ligand_Perturbations_down',
 'LINCS_L1000_Ligand_Perturbations_up',
 'Ligand_Perturbations_from_GEO_down',
 'Ligand_Perturbations_from_GEO_up',
 'MAGMA_Drugs_and_Diseases',
 'MAGNET_2023',
 'MCF7_Perturbations_from_GEO_down',

'MCF7_Perturbations_from_GEO_up',
 'MGI_Mammalian_Phenotype_Level_4_2021',
 'MGI_Mammalian_Phenotype_Level_4_2024',
 'MSigDB_Computational',
 'MSigDB_Hallmark_2020',
 'MSigDB_Oncogenic_Signatures',
 'Metabolomics_Workbench_Metabolites_2022',
 'Microbe_Perturbations_from_GEO_down',
 'Microbe_Perturbations_from_GEO_up',
 'MoTrPAC_2023',
 'Mouse_Gene_Atlas',
 'NCI-60_Cancer_Cell_Lines',
 'NCI-Nature_2016',
 'NIBR_DRUGseq_2025_down',
 'NIBR_DRUGseq_2025_up',
 'NURSA_Human_Endogenous_Complexome',
 'OMIM_Disease',
 'OMIM_Expanded',
 'Old_CMAP_down',
 'Old_CMAP_up',
 'Orphanet_Augmented_2021',
 'PFOCR_Pathways_2023',
 'PPI_Hub_Proteins',
 'PanglaoDB_Augmented_2021',
 'Panther_2015',
 'Panther_2016',
 'PerturbAtlas',
 'PerturbAtlas_MouseGenePerturbationSigs',
 'PerturbSeq_ReplogleK562',
 'PerturbSeq_ReplogleRPE1',
 'Pfam_Domains_2019',
 'Pfam_InterPro_Domains',
 'PheWeb_2019',
 'PhenGenI_Association_2021',
 'Phosphatase_Substrates_from_DEPOD',
 'ProteomicsDB_2020',
 'Proteomics_Drug_Atlas_2023',
 'RNA-Seq_Disease_Gene_and_Drug_Signatures_from_GEO',
 'Rare_Diseases_AutoRIF_ARCHS4_Predictions',
 'Rare_Diseases_AutoRIF_Gene_Lists',
 'Rare_Diseases_GeneRIF_ARCHS4_Predictions',
 'Rare_Diseases_GeneRIF_Gene_Lists',
 'Reactome_2022',
 'Reactome_Pathways_2024',
 'RummaGEO_DrugPerturbations_2025',
 'RummaGEO_GenePerturbations_2025',
 'Rummagene_kinases',

```

'Rummagene_signatures',
'Rummagene_transcription_factors',
'SILAC_Phosphoproteomics',
'SubCell_BarCode',
'SynGO_2022',
'SynGO_2024',
'SysMyo_Muscle_Gene_Sets',
'TF-LOF_Expression_from_GEO',
'TF_Perturbations_Followed_by_Expression',
'TG_GATES_2020',
'TISSUES_Curated_2025',
'TISSUES_Experimental_2025',
'TRANSFAC_and_JASPAR_PWMs',
'TRRUST_Transcription_Factors_2019',
'Table_Mining_of_CRISPR_Studies',
'Tabula_Muris',
'Tabula_Sapiens',
'TargetScan_microRNA',
'TargetScan_microRNA_2017',
'The_Kinase_Library_2023',
'The_Kinase_Library_2024',
'Tissue_Protein_Expression_from_Human_Proteome_Map',
'Tissue_Protein_Expression_from_ProteomicsDB',
'Transcription_Factor_PPIs',
'UK_Biobank_GWAS_v1',
'Virus-Host_PPI_P-HIPSTer_2020',
'VirusMINT',
'Virus_Perturbations_from_GEO_down',
'Virus_Perturbations_from_GEO_up',
'WikiPathway_2021_Human',
'WikiPathway_2023_Human',
'WikiPathways_2013',
'WikiPathways_2015',
'WikiPathways_2016',
'WikiPathways_2019_Human',
'WikiPathways_2019_Mouse',
'WikiPathways_2024_Human',
'WikiPathways_2024_Mouse',
'dbGaP',
'huMAP',
'lncHUB_lncRNA_Co-Expression',
'miRTarBase_2017']

```

```

[32]: pre_res = gp.prerank(rnk = ranking,
                        gene_sets = 'WikiPathways_2024_Human', # Wiki geneset.
                        seed = 42)

```

```
[35]: pre_res.res2d.to_csv('supplemental figure x/GSEA/macro_nf2_wikipathways.csv')
```

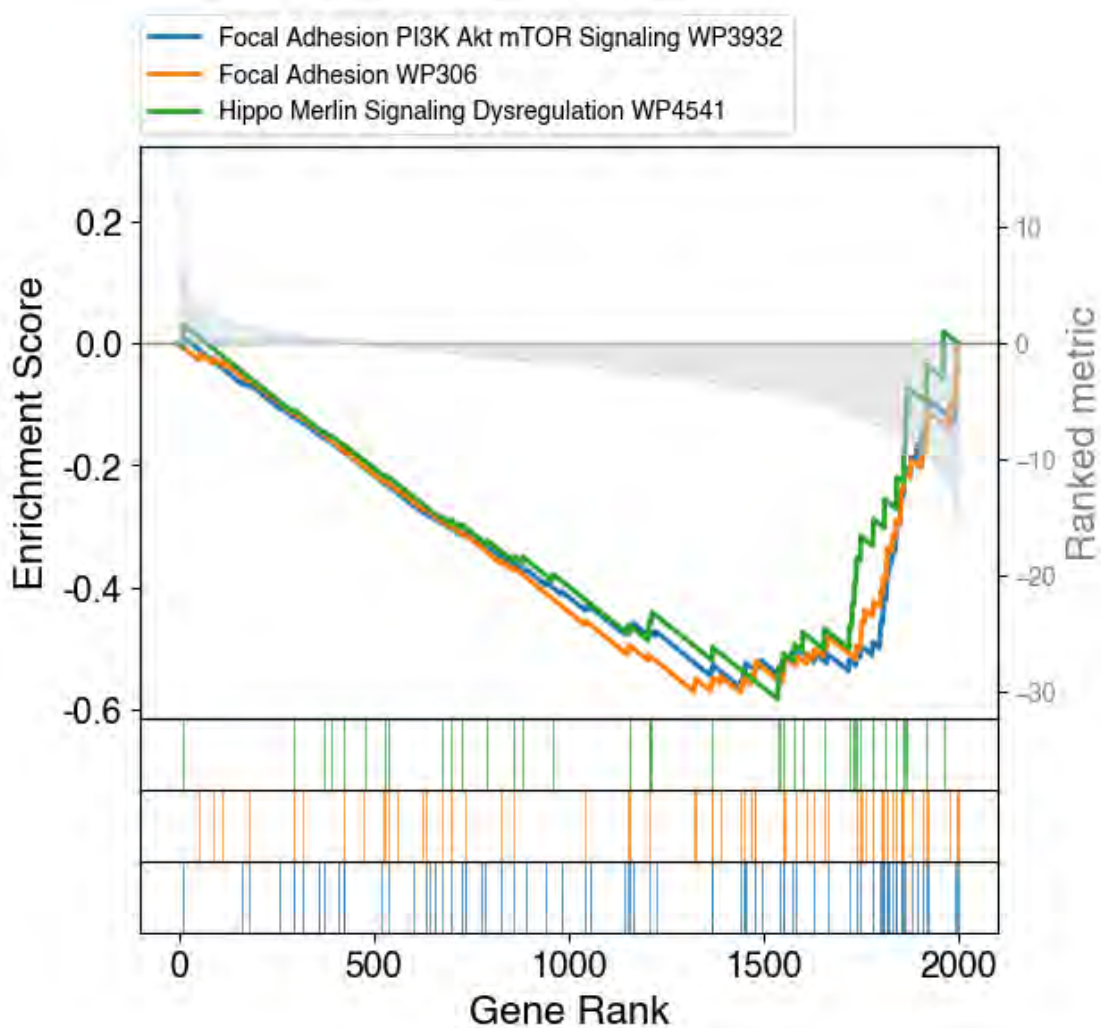
```
[40]: terms = pre_res.res2d.Term
selected_terms = [terms[i] for i in [2,3,4]] # get the index positions from
↳ pre_res.res2d

axs = pre_res.plot(terms=selected_terms,
                    #legend_kws={'loc': (1.2, 0)}, # set the legend loc
                    show_ranking=True, # whether to show the second yaxis
                    figsize=(3,4))

axs.savefig('supplemental figure x/GSEA/NF2_macrophage_wiki.pdf',
↳ bbox_inches='tight' )
```

1 extra bytes in post.stringData array

'created' timestamp seems very low; regarding as unix timestamp



8.1.2 Custom gene set for M1/M2 macrophages using Supplemental Table S2B from Xue et al. 2014, Immunity

```
[ ]: # Save the genes in two rows as tab separated values (.txt)' read in with
↳BBedit and save as .gmt

pre_res_custom = gp.prerank(rnk=ranking,
                             gene_sets='supplemental figure x/GSEA/custom_m1m2_marker.
↳gmt',
                             seed=42)
```

```
[217]: pre_res_custom.results
```

```
[217]: {'M1_markers': {'name': 'prerank',
  'es': -0.5073266231424559,
  'nes': -1.2810298089345398,
  'pval': 0.09837728194726167,
  'fdr': 0.17474747474747473,
  'fwerp': 0.16616616616616617,
  'tag %': '12/30',
  'gene %': '20.80%',
  'lead_genes':
  'CNKSR3;IFIT1;CSF1;IFI27;IFIT3;MX1;HSPA1A;IFI6;DNAJB1;IFIT2;ISG15;IFITM3',
  'matched_genes': 'MX2;DKK2;EPSTI1;CPNE8;CXCL9;SERPINH1;SERPINE1;ITGA5;DYSF;OAS
  L;CXCL10;CCL8;RSAD2;KCNJ2;MID1;GBP1;HSPA1B;WARS;IFITM3;ISG15;IFIT2;DNAJB1;IFI6;H
  SPA1A;MX1;IFIT3;IFI27;CSF1;IFIT1;CNKSR3',
  'hits': [97,
    222,
    241,
    338,
    363,
    400,
    517,
    536,
    781,
    790,
    882,
    1065,
    1174,
    1175,
    1248,
    1326,
    1406,
    1456,
    1585,
```

1624,
1659,
1664,
1670,
1679,
1725,
1761,
1803,
1808,
1820,
1900],

'RES': [-0.0005076142131979696,
-0.0010152284263959391,
-0.0015228426395939086,
-0.0020304568527918783,
-0.002538071065989848,
-0.0030456852791878176,
-0.0035532994923857873,
-0.0040609137055837565,
-0.004568527918781726,
-0.005076142131979696,
-0.005583756345177666,
-0.006091370558375635,
-0.006598984771573605,
-0.007106598984771575,
-0.007614213197969544,
-0.008121827411167513,
-0.008629441624365483,
-0.009137055837563452,
-0.009644670050761422,
-0.010152284263959392,
-0.010659898477157361,
-0.011167512690355331,
-0.0116751269035533,
-0.01218274111675127,
-0.01269035532994924,
-0.01319796954314721,
-0.01370558375634518,
-0.01421319796954315,
-0.014720812182741119,
-0.015228426395939089,
-0.015736040609137057,
-0.016243654822335026,
-0.016751269035532996,
-0.017258883248730966,
-0.017766497461928935,
-0.018274111675126905,

-0.018781725888324875,
-0.019289340101522844,
-0.019796954314720814,
-0.020304568527918784,
-0.020812182741116753,
-0.021319796954314723,
-0.021827411167512693,
-0.022335025380710662,
-0.022842639593908632,
-0.0233502538071066,
-0.02385786802030457,
-0.02436548223350254,
-0.02487309644670051,
-0.02538071065989848,
-0.02588832487309645,
-0.02639593908629442,
-0.02690355329949239,
-0.02741116751269036,
-0.02791878172588833,
-0.0284263959390863,
-0.028934010152284268,
-0.029441624365482238,
-0.029949238578680207,
-0.030456852791878177,
-0.030964467005076147,
-0.03147208121827411,
-0.03197969543147208,
-0.03248730964467005,
-0.03299492385786802,
-0.03350253807106599,
-0.03401015228426396,
-0.03451776649746193,
-0.0350253807106599,
-0.03553299492385787,
-0.03604060913705584,
-0.03654822335025381,
-0.03705583756345178,
-0.03756345177664975,
-0.03807106598984772,
-0.03857868020304569,
-0.03908629441624366,
-0.03959390862944163,
-0.0401015228426396,
-0.04060913705583757,
-0.04111675126903554,
-0.04162436548223351,
-0.042131979695431476,

-0.042639593908629446,
-0.043147208121827416,
-0.043654822335025385,
-0.044162436548223355,
-0.044670050761421325,
-0.045177664974619294,
-0.045685279187817264,
-0.046192893401015234,
-0.0467005076142132,
-0.04720812182741117,
-0.04771573604060914,
-0.04822335025380711,
-0.04873096446700508,
-0.04923857868020305,
-0.026677388525249188,
-0.027185002738447157,
-0.027692616951645127,
-0.028200231164843097,
-0.028707845378041066,
-0.029215459591239036,
-0.029723073804437006,
-0.030230688017634975,
-0.030738302230832945,
-0.031245916444030915,
-0.031753530657228884,
-0.032261144870426854,
-0.032768759083624824,
-0.033276373296822793,
-0.03378398751002076,
-0.03429160172321873,
-0.0347992159364167,
-0.03530683014961467,
-0.03581444436281264,
-0.03632205857601061,
-0.03682967278920858,
-0.03733728700240655,
-0.03784490121560452,
-0.03835251542880249,
-0.03886012964200046,
-0.03936774385519843,
-0.0398753580683964,
-0.04038297228159437,
-0.04089058649479234,
-0.04139820070799031,
-0.04190581492118828,
-0.04241342913438625,
-0.04292104334758422,

-0.04342865756078219,
-0.04393627177398016,
-0.044443885987178126,
-0.044951500200376096,
-0.045459114413574066,
-0.045966728626772035,
-0.046474342839970005,
-0.046981957053167975,
-0.047489571266365944,
-0.047997185479563914,
-0.048504799692761884,
-0.049012413905959853,
-0.04952002811915782,
-0.05002764233235579,
-0.05053525654555376,
-0.05104287075875173,
-0.0515504849719497,
-0.05205809918514767,
-0.05256571339834564,
-0.05307332761154361,
-0.05358094182474158,
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-0.27596797771207376,
...]]}

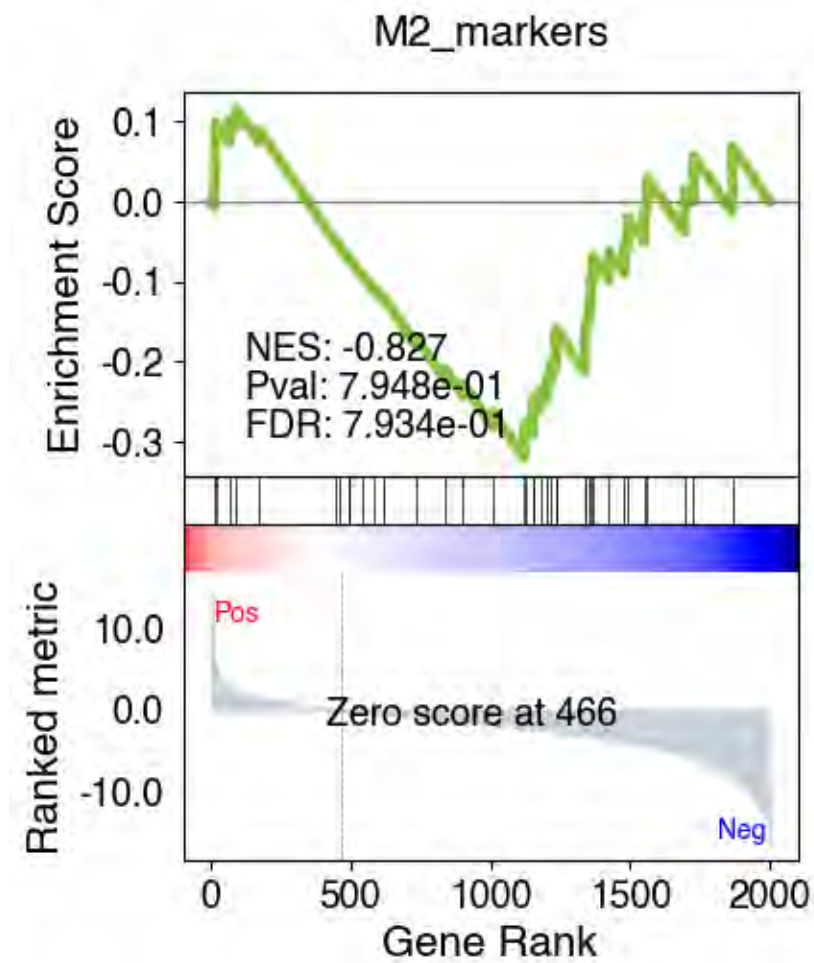
```

```

[218]: pre_res_custom.res2d.to_csv('supplemental figure x/GSEA/
      ↪macro_deg_pre_res_NF2_custom.res2d.csv')

```

```
[220]: terms = pre_res_custom.res2d.Term
      axs = pre_res_custom.plot(terms=terms[1])
```



```
[223]: selected_terms = [terms[i] for i in [0,1]] # get the index positions from
      ↪ pre_res.res2d

      axs = pre_res_custom.plot(terms=selected_terms,
                                #legend_kws={'loc': (1.2, 0)}, # set the legend loc
                                show_ranking=True, # whether to show the second yaxis
                                figsize=(3,4))

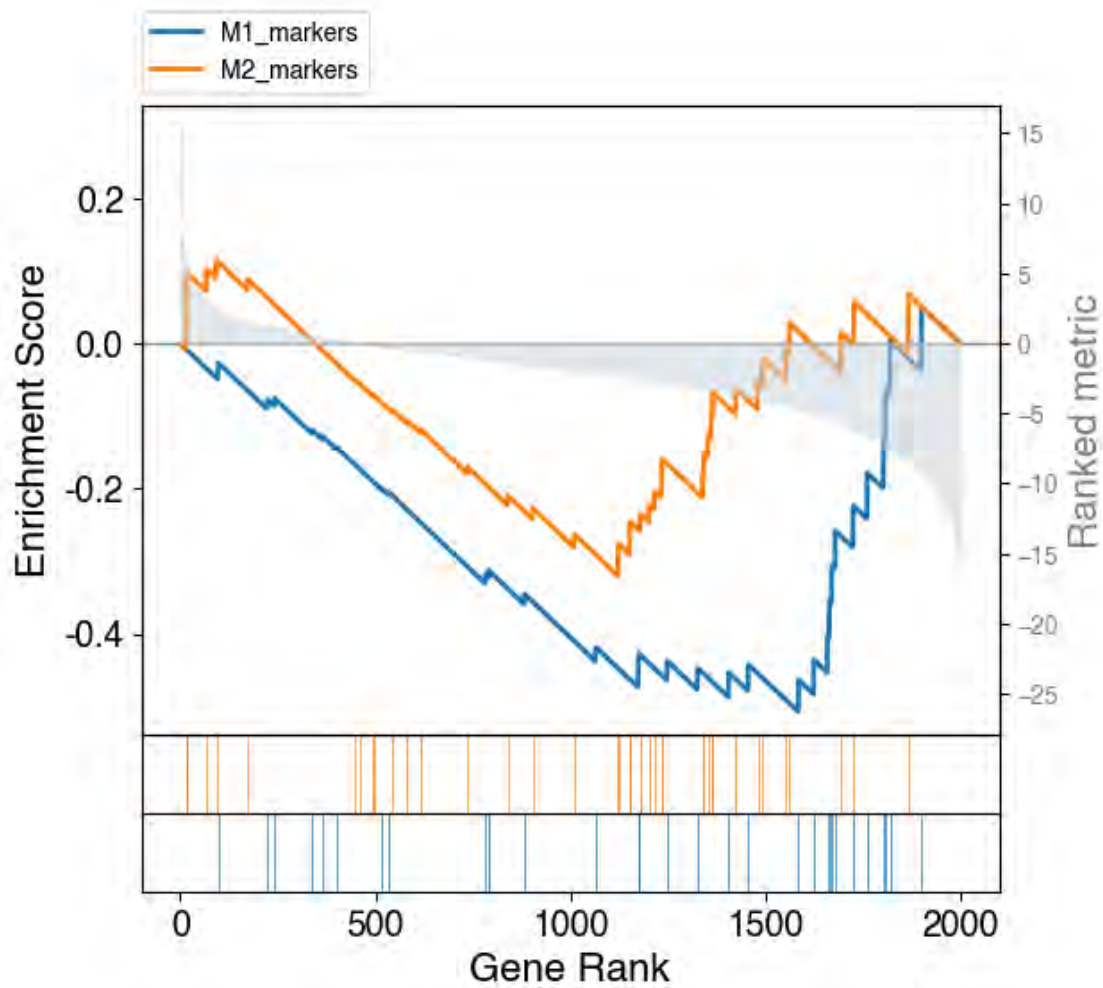
      axs.savefig('supplemental figure x/GSEA/NF2_macrophage_gsea_custom.pdf',
      ↪ bbox_inches='tight' )
```

1 extra bytes in post.stringData array
 'created' timestamp seems very low; regarding as unix timestamp
 Zapf NOT subset; don't know how to subset; dropped

```

feat NOT subset; don't know how to subset; dropped
meta NOT subset; don't know how to subset; dropped
morx NOT subset; don't know how to subset; dropped

```



```

[100]: ## Plot overall Integrated object colored by NF2 vs Sporadic
fig, ((ax1, ax2), (ax3, ax4)) = plt.subplots(nrows=2, ncols=2, figsize=(10, 10),
↳ dpi=600, )
ax1_grey = sc.pl.umap(integrated,
                        use_raw = False,
                        frameon=False,
                        show=False,
                        ax = ax1,
                        # color = "pred.type",
                        s = 30,
                        # palette = ct_dict,
                        # edgecolor = 'black',

```

```

        # linewidth= 0.5,
        alpha = 0.9,
    )
ax1_dict = sc.pl.umap(schwann_nf2,
    use_raw = False,
    frameon=False,
    show=False,
    ax = ax1,
    color = "orig.ident",
    s = 25,
    palette = color_dict,
    edgecolor = 'black',
    linewidth= 0.4,
    alpha = 0.9,
)
ax2_grey = sc.pl.umap(integrated,
    use_raw = False,
    frameon=False,
    show=False,
    ax = ax2,
    # color = "pred.type",
    s = 30,
    # palette = ct_dict,
    # edgecolor = 'black',
    # linewidth= 0.5,
    alpha = 0.9,
)
ax1_dict = sc.pl.umap(schwann_sporadic,
    use_raw = False,
    frameon=False,
    show=False,
    ax = ax2,
    color = "orig.ident",
    s = 25,
    palette = color_dict,
    edgecolor = 'black',
    linewidth= 0.4,
    alpha = 0.9,
)
ax3_grey = sc.pl.umap(integrated,
    use_raw = False,
    frameon=False,
    show=False,
    ax = ax3,
    # color = "pred.type",
    s = 30,
    # palette = ct_dict,

```

```

        # edgecolor = 'black',
        # linewidth= 0.5,
        alpha = 0.9,
    )
ax3_dict = sc.pl.umap(macro_nf2,
                      use_raw = False,
                      frameon=False,
                      show=False,
                      ax = ax3,
                      color = "orig.ident",
                      s = 25,
                      palette = color_dict,
                      edgecolor = 'black',
                      linewidth= 0.4,
                      alpha = 0.9,
                      )
ax4_grey = sc.pl.umap(integrated,
                      use_raw = False,
                      frameon=False,
                      show=False,
                      ax = ax4,
                      # color = "pred.type",
                      s = 30,
                      # palette = ct_dict,
                      # edgecolor = 'black',
                      # linewidth= 0.5,
                      alpha = 0.9,
                      )
ax4_dict = sc.pl.umap(macro_sporadic,
                      use_raw = False,
                      frameon=False,
                      show=False,
                      ax = ax4,
                      color = "orig.ident",
                      s = 25,
                      palette = color_dict,
                      edgecolor = 'black',
                      linewidth= 0.4,
                      alpha = 0.9,
                      )

handles, labels = plt.gca().get_legend_handles_labels()

leg1 = ax1.legend(prop = {'size':'14'}, markerscale=1.6, frameon=False,
                 ↪ bbox_to_anchor=(0, 1))
for lh in leg1.legend_handles:

```

```

        lh.set_alpha(0.9)
        lh.set_edgecolor("black")

leg2 = ax2.legend(prop = {'size':'14'}, markerscale=1.6, frameon=False,
↳bbox_to_anchor=(1, 1))
for lh in leg2.legend_handles:
    lh.set_alpha(0.9)
    lh.set_edgecolor("black")

leg3 = ax3.legend(prop = {'size':'14'}, markerscale=1.6, frameon=False,
↳bbox_to_anchor=(0, 1))
for lh in leg3.legend_handles:
    lh.set_alpha(0.9)
    lh.set_edgecolor("black")

leg4 = ax4.legend(prop = {'size':'14'}, markerscale=1.6, frameon=False,
↳bbox_to_anchor=(1, 1))
for lh in leg4.legend_handles:
    lh.set_alpha(0.9)
    lh.set_edgecolor("black")

ax1.set_title("NF2 Schwann cells")
ax2.set_title("Sporadic Schwann cells")
ax3.set_title("NF2 Macrophages")
ax4.set_title("Sporadic Macrophages")

fig.savefig('supplemental figure x/cells_per_type.pdf',bbox_inches='tight')
fig.savefig('supplemental figure x/cells_per_type.png',bbox_inches='tight')

```

```

/Users/chittiboinap/anaconda3/envs/spatial/lib/python3.9/site-
packages/scanpy/plotting/_tools/scatterplots.py:394: UserWarning: No data for
colormapping provided via 'c'. Parameters 'cmap', 'norm' will be ignored
    cax = scatter(
/Users/chittiboinap/anaconda3/envs/spatial/lib/python3.9/site-
packages/scanpy/plotting/_tools/scatterplots.py:1234: FutureWarning: The default
value of 'ignore' for the `na_action` parameter in pandas.Categorical.map is
deprecated and will be changed to 'None' in a future version. Please set
na_action to the desired value to avoid seeing this warning
    color_vector = pd.Categorical(values.map(color_map))
/Users/chittiboinap/anaconda3/envs/spatial/lib/python3.9/site-
packages/scanpy/plotting/_tools/scatterplots.py:394: UserWarning: No data for
colormapping provided via 'c'. Parameters 'cmap' will be ignored
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/Users/chittiboinap/anaconda3/envs/spatial/lib/python3.9/site-
packages/scanpy/plotting/_tools/scatterplots.py:394: UserWarning: No data for
colormapping provided via 'c'. Parameters 'cmap', 'norm' will be ignored

```

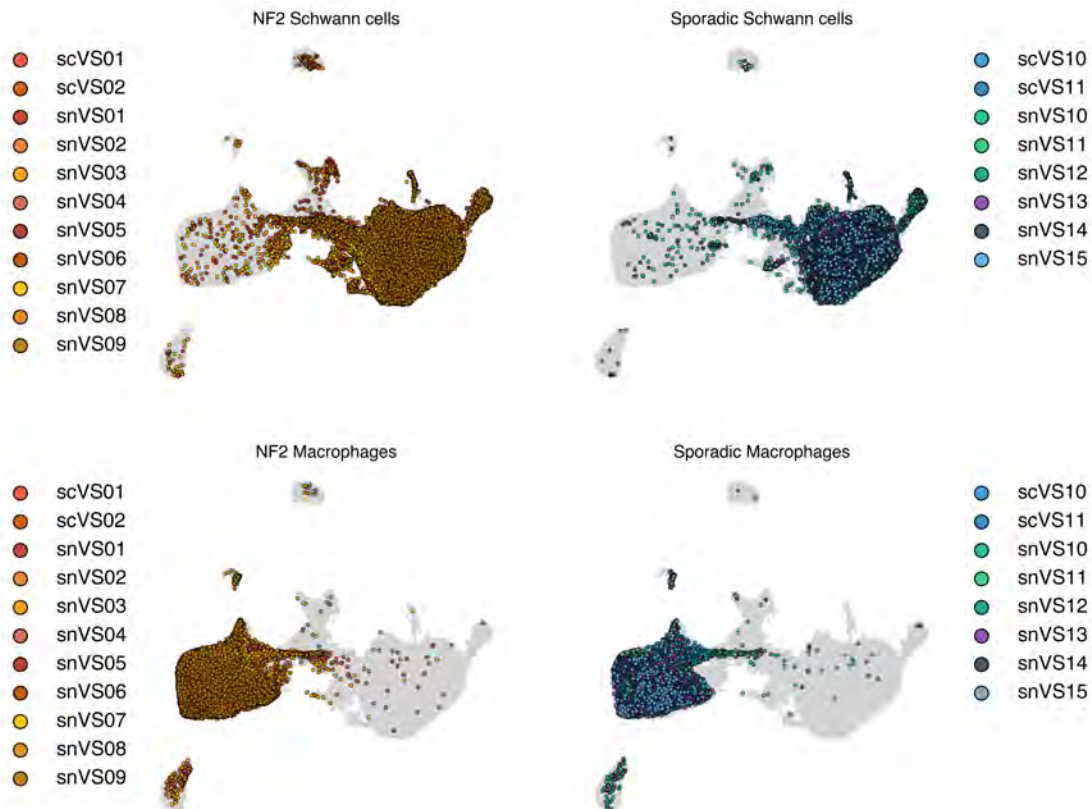
```

cax = scatter(
/Users/chittiboinap/anaconda3/envs/spatial/lib/python3.9/site-
packages/scanpy/plotting/_utils.py:432: ImplicitModificationWarning: Trying to
modify attribute `._uns` of view, initializing view as actual.
adata.uns[value_to_plot + '_colors'] = colors_list
/Users/chittiboinap/anaconda3/envs/spatial/lib/python3.9/site-
packages/scanpy/plotting/_tools/scatterplots.py:1234: FutureWarning: The default
value of 'ignore' for the `na_action` parameter in pandas.Categorical.map is
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/Users/chittiboinap/anaconda3/envs/spatial/lib/python3.9/site-
packages/scanpy/plotting/_tools/scatterplots.py:394: UserWarning: No data for
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cax = scatter(
/Users/chittiboinap/anaconda3/envs/spatial/lib/python3.9/site-
packages/scanpy/plotting/_utils.py:432: ImplicitModificationWarning: Trying to
modify attribute `._uns` of view, initializing view as actual.
adata.uns[value_to_plot + '_colors'] = colors_list
/Users/chittiboinap/anaconda3/envs/spatial/lib/python3.9/site-
packages/scanpy/plotting/_tools/scatterplots.py:1234: FutureWarning: The default
value of 'ignore' for the `na_action` parameter in pandas.Categorical.map is
deprecated and will be changed to 'None' in a future version. Please set
na_action to the desired value to avoid seeing this warning
color_vector = pd.Categorical(values.map(color_map))
/Users/chittiboinap/anaconda3/envs/spatial/lib/python3.9/site-

```

```
packages/scanpy/plotting/_tools/scatterplots.py:394: UserWarning: No data for
colormapping provided via 'c'. Parameters 'cmap' will be ignored
```

```
cax = scatter(
1 extra bytes in post.stringData array
'created' timestamp seems very low; regarding as unix timestamp
Zapf NOT subset; don't know how to subset; dropped
feat NOT subset; don't know how to subset; dropped
meta NOT subset; don't know how to subset; dropped
morx NOT subset; don't know how to subset; dropped
```



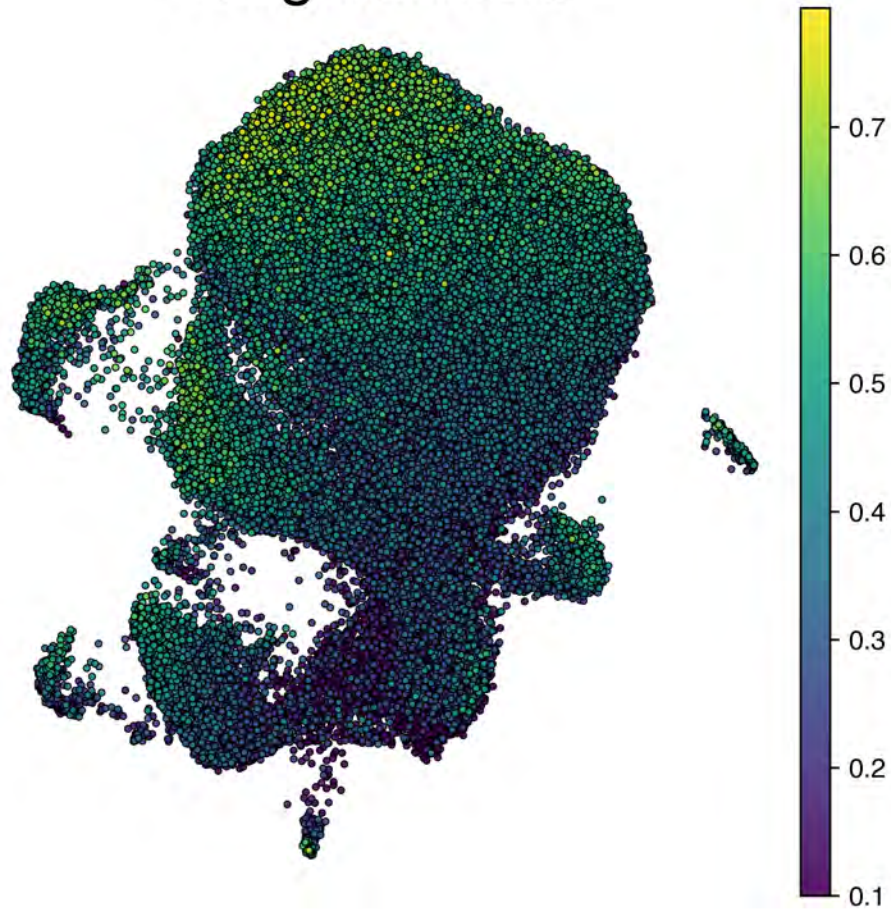
9 NF2 Depletion score and VEGFA Expression

```
[16]: ...
00
fig, ax1 = plt.subplots(nrows=1,ncols=1,figsize=(6,6), dpi=600)
ax1_dict = sc.pl.umap(schwann,
                      use_raw = False,
                      frameon=False,
                      show=False,
                      ax = ax1,
```

```
color = "score21",  
s = 25,  
cmap='viridis',  
edgecolor = 'black',  
linewidth= 0.5,  
alpha = 0.9,  
vmin=0.1,  
# vmax=5  
)
```

```
ax1.set_title("Original MDS", fontweight="bold", fontsize=24)  
fig.savefig('fig_2_new/schwann_old_mds.png', bbox_inches='tight')
```

Original MDS



```
[35]: 'FGF2' in new_mds
```

```
[35]: True
```

9.1 Save Schwann dataset with new_mds

```
[23]: schwann.raw.var.rename(columns={'_index': 'features'}, inplace=True)
      schwann.write_h5ad('fig_2_new/schwann_new_mds.h5ad')

[144]: ## Create "blended" score which is a function of VEGFA expression and NF2
      ↳ Depletion score

      # create new column "NF2_status". Cells below cutoff (no NF2 depletion) will be
      ↳ annotated as 0
      def NF2_status(row):
          if row['score1'] >= 0.1:
              return row['score1']
          elif row['score1'] < 0.1:
              return 0

      schwann.obs['NF2_status'] = schwann.obs.apply(NF2_status, axis=1)

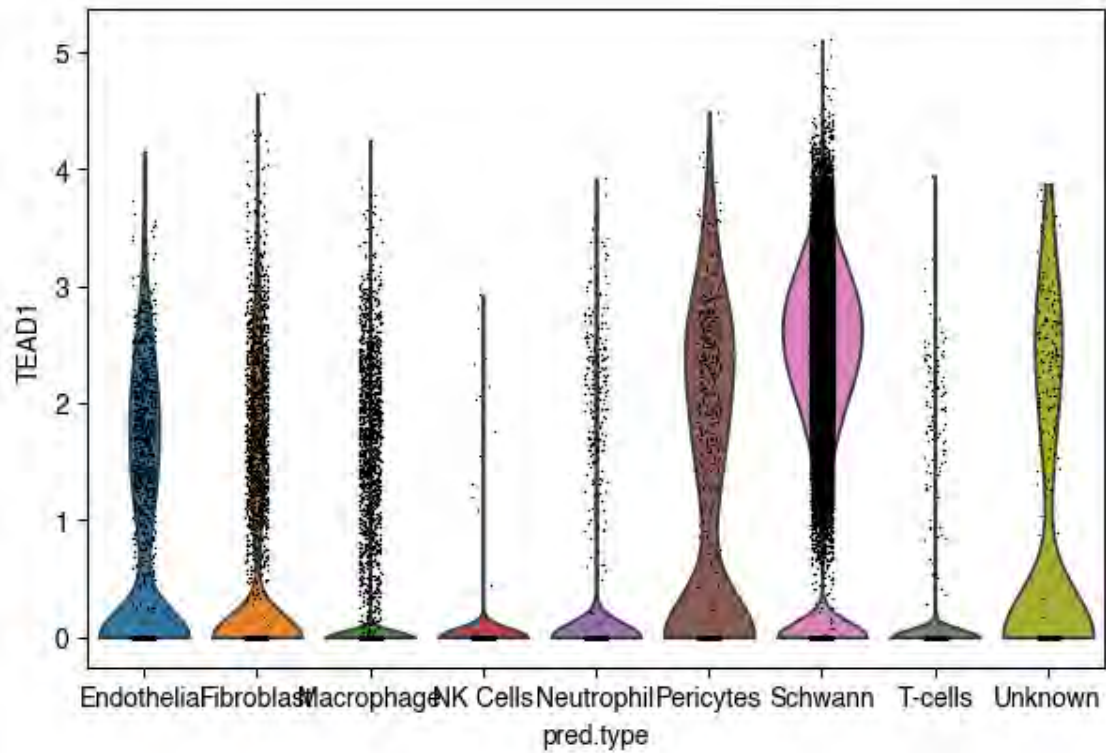
      # create new column "VEGFA_status". Cells below cutoff (low VEGFA expression)
      ↳ will be annotated as 0
      def VEGFA_status(x):
          if x >= 1:
              return x
          elif x < 1:
              return 0

      schwann.obs['VEGFA_status'] = sc.get.obs_df(schwann, 'VEGFA').
      ↳ apply(VEGFA_status)

      # NF2 depletion * VEGFA expression - cells with low NF2 depletion and/or low
      ↳ VEGFA expression will return 0
      schwann.obs['NF2_VEGFA_status'] = schwann.obs['NF2_status'] * schwann.
      ↳ obs['VEGFA_status'] #may adjust funtion
```

9.2 TEAD distribution per cell

```
[4]: sc.pl.violin(integrated, keys='TEAD1', groupby='pred.type', use_raw=False,
                  #stripplot=True, jitter=0.4, size=1
                  )
```



10 Within patient comparisons

```
[13]: integrated.obs['orig.ident'].value_counts()
```

```
[13]: orig.ident
snVS03    17566
snVS04    11722
snVS07     9536
snVS05     7230
snVS02     6855
scVS02     6578
scVS11     5963
snVS08     5875
snVS11     3330
snVS09     2945
snVS14     2934
snVS12     2793
snVS13     2597
snVS01     1775
scVS10     1601
scVS01     1358
```

```
snVS10      1182
snVS15       904
snVS06       686
Name: count, dtype: int64
```

```
[15]: pt5 = integrated[integrated.obs['orig.ident'].isin(['snVS05', 'snVS08'])]
      pt8 = integrated[integrated.obs['orig.ident'].isin(['snVS07', 'snVS09'])]
      pt5.obs['orig.ident'].value_counts()
```

```
[15]: orig.ident
snVS05      7230
snVS08      5875
Name: count, dtype: int64
```

```
[16]: sc.tl.rank_genes_groups(pt5, groupby='orig.ident', use_raw=False,
    ↪key_added='rank_genes_groups_pt5')
sc.tl.rank_genes_groups(pt8, groupby='orig.ident', use_raw=False,
    ↪key_added='rank_genes_groups_pt8')

pt5_schwann = pt5[pt5.obs['pred.type'] == 'Schwann']
pt8_schwann = pt8[pt8.obs['pred.type'] == 'Schwann']

sc.tl.rank_genes_groups(pt5_schwann, groupby='orig.ident', use_raw=False,
    ↪key_added='rank_genes_groups_pt5_schwann')
sc.tl.rank_genes_groups(pt8_schwann, groupby='orig.ident', use_raw=False,
    ↪key_added='rank_genes_groups_pt8_schwann')
```

```
/opt/anaconda3/envs/singlecell/lib/python3.14/site-
packages/scanpy/_utils/__init__.py:472: ImplicitModificationWarning: Trying to
modify attribute `.obs` of view, initializing view as actual.
adata._sanitize()
/opt/anaconda3/envs/singlecell/lib/python3.14/site-
packages/scanpy/_utils/__init__.py:472: ImplicitModificationWarning: Trying to
modify attribute `.obs` of view, initializing view as actual.
adata._sanitize()
/opt/anaconda3/envs/singlecell/lib/python3.14/site-
packages/scanpy/_utils/__init__.py:472: ImplicitModificationWarning: Trying to
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adata._sanitize()
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/opt/anaconda3/envs/singlecell/lib/python3.14/site-
packages/scanpy/_utils/__init__.py:472: ImplicitModificationWarning: Trying to
modify attribute `.obs` of view, initializing view as actual.
adata._sanitize()
/opt/anaconda3/envs/singlecell/lib/python3.14/site-
```

```

packages/scanpy/_utils/__init__.py:472: ImplicitModificationWarning: Trying to
modify attribute `.obs` of view, initializing view as actual.
    adata._sanitize()
/opt/anaconda3/envs/singlecell/lib/python3.14/site-
packages/scanpy/_utils/__init__.py:472: ImplicitModificationWarning: Trying to
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    adata._sanitize()
/opt/anaconda3/envs/singlecell/lib/python3.14/site-
packages/scanpy/_utils/__init__.py:472: ImplicitModificationWarning: Trying to
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    adata._sanitize()
/opt/anaconda3/envs/singlecell/lib/python3.14/site-
packages/scanpy/_utils/__init__.py:472: ImplicitModificationWarning: Trying to
modify attribute `.obs` of view, initializing view as actual.
    adata._sanitize()
/opt/anaconda3/envs/singlecell/lib/python3.14/site-
packages/scanpy/_utils/__init__.py:472: ImplicitModificationWarning: Trying to
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/opt/anaconda3/envs/singlecell/lib/python3.14/site-
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    adata._sanitize()
/opt/anaconda3/envs/singlecell/lib/python3.14/site-
packages/scanpy/_utils/__init__.py:472: ImplicitModificationWarning: Trying to
modify attribute `.obs` of view, initializing view as actual.
    adata._sanitize()
/opt/anaconda3/envs/singlecell/lib/python3.14/site-
packages/scanpy/_utils/__init__.py:472: ImplicitModificationWarning: Trying to
modify attribute `.obs` of view, initializing view as actual.
    adata._sanitize()
/opt/anaconda3/envs/singlecell/lib/python3.14/site-
packages/scanpy/_utils/__init__.py:472: ImplicitModificationWarning: Trying to
modify attribute `.obs` of view, initializing view as actual.
    adata._sanitize()
/opt/anaconda3/envs/singlecell/lib/python3.14/site-

```

packages/scanpy/_utils/_init_.py:472: ImplicitModificationWarning: Trying to modify attribute `obs` of view, initializing view as actual.

```
adata._sanitize()
```

/opt/anaconda3/envs/singlecell/lib/python3.14/site-

packages/scanpy/tools/_rank_genes_groups.py:829: ImplicitModificationWarning: Trying to modify attribute `uns` of view, initializing view as actual.

```
adata.uns[key_added] = {}
```

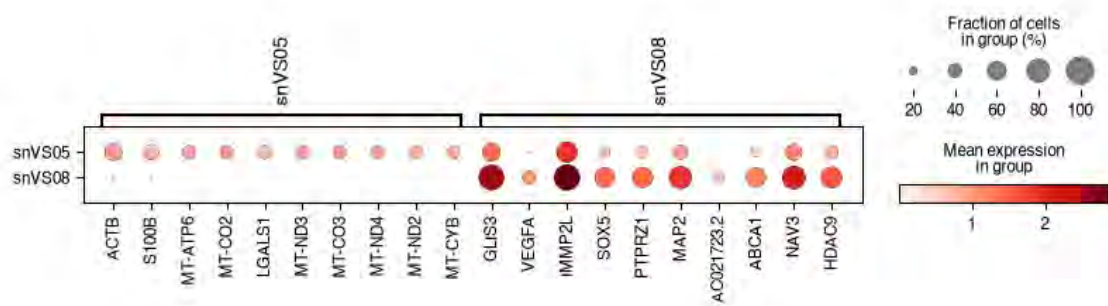
/opt/anaconda3/envs/singlecell/lib/python3.14/site-

packages/scanpy/tools/_rank_genes_groups.py:829: ImplicitModificationWarning: Trying to modify attribute `uns` of view, initializing view as actual.

```
adata.uns[key_added] = {}
```

```
[18]: sc.pl.rank_genes_groups_dotplot(pt5_schwann,
    ↪key='rank_genes_groups_pt5_schwann', use_raw=False)
sc.pl.rank_genes_groups_dotplot(pt8_schwann,
    ↪key='rank_genes_groups_pt8_schwann', use_raw=False)
```

WARNING: Dendrogram not added. Dendrogram is added only when the number of categories to plot > 2



WARNING: Dendrogram not added. Dendrogram is added only when the number of categories to plot > 2

